# MODASYS

## Model for Decision-Making in Agri-Environmental Systems




## A Quick User Guide




HOLOSEU

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# 1. Introduction

MODASYS (Model for Decision-Making in Agri-Environmental Systems) is a whole-farm, systems-based digital platform developed in the HOLOSEU project. It supports agricultural land-use planning, decision-making for climate change mitigation and adaptation, and national inventory reporting. MODASYS estimates greenhouse gas (GHG) emissions of carbon dioxide ($CO_2$), nitrous oxide ($N_2O$) and methane ($CH_4$), soil organic carbon change, carbon sequestration in trees, life-cycle (upstream and downstream) energy emissions, soil health and biodiversity indices, production, economics, and the farm's total and net carbon balance.

MODASYS is the latest in a series of agricultural modelling platforms. HOLOS-IE was developed for Irish farming systems at University College Dublin (UCD), and MODASYS extends that work to European Union agricultural contexts. HOLOS-IE was developed through Research Ireland (formerly Science Foundation Ireland)'s National Challenge Fund (Future Digital Challenge), in collaboration with the ReLive project funded by Transnational ERA-NET (EU) through the Department of Agriculture, Food and the Marine, Ireland. The HOLOSEU project, which builds MODASYS, is supported through ICT-AGRI-FOOD with national co-funding from 1 January 2025. It aims to enable harmonised assessment and comparison of agricultural GHG emissions and mitigation strategies across EU member states, and adds modules on agrobiodiversity, the bioeconomy and circularity.

The model lets farmers, advisors, policymakers and researchers evaluate existing practices and explore alternative management that sustains productivity while reducing the farm's environmental footprint. A farm is built from components (fields of crops and grass, livestock herds, agroforestry systems and farm infrastructure), so the same tool covers a single land parcel, a mono-enterprise farm or a mixed farming system.

**N.B.:** MODASYS has been primarily designed to encourage stakeholders, including farmers, to explore existing and alternative methods for improving production, mitigating greenhouse gas emissions and addressing environmental pollution. It also supports carbon balance accounting and cost–benefit analysis at farm-to-national level across European Union member states. The estimates are approximate and may change significantly as new scientific findings emerge and the model is further calibrated and validated. University College Dublin (UCD) and the collaborating institutes do not warrant the reliability of this tool for uses other than research and education at this stage. Users should consult qualified agricultural advisors for farm-level decisions.

This guide walks through the application using **Dairy-Ire-F1**, an example Irish dairy farm in Co. Kildare (soil polygon 1352) with a grass field, a crop field, an agroforestry system, a dairy herd and farm infrastructure (anaerobic digestion, buildings and annual energy flows). All screenshots were taken from the running application.

# 2. Getting Started

## 2.1 Software

MODASYS can be obtained from the HOLOSEU project website (www.holoseu.net) or the MODASYS website (www.modasys.org), or by contacting the Project Lead (ibrahim.khalil1@ucd.ie). It is free to use for research and education.

## 2.2 Installation and First Launch

- Run the MODASYS **setup** installer.
- Launch **MODASYS** from the Windows Start menu (search "MODASYS").
- MODASYS opens with the copyright, licensing and disclaimer page (Figure 2.1). Use the **Language** drop-down in the top right to choose **English**, **French** or **Polish**; the whole interface switches immediately. Read the notices and click **OK**.
- A short loading screen appears while the model's data tables are prepared.

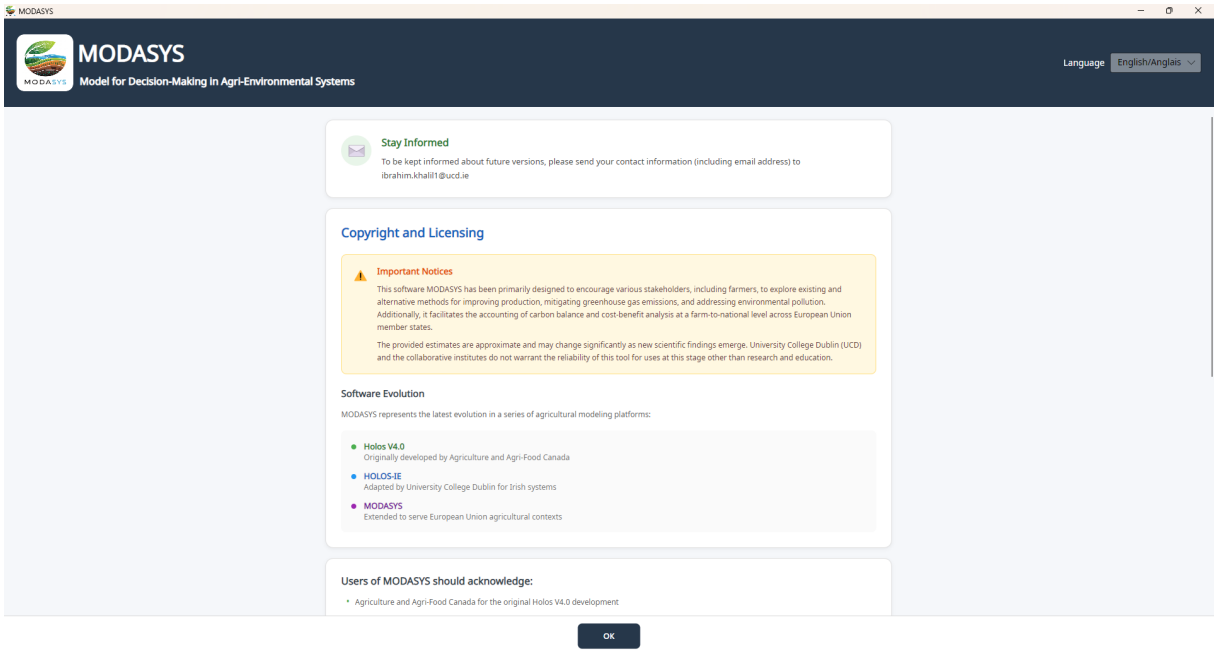



**Figure 2.1:** *The MODASYS start-up page with the copyright and licensing notices. The language selector is in the top right; click OK at the bottom to continue.*

## 2.3 The Start Screen

After loading, MODASYS asks whether you want to create a new farm, open a farm you have worked on before, or import a farm from a file (Figure 2.2).

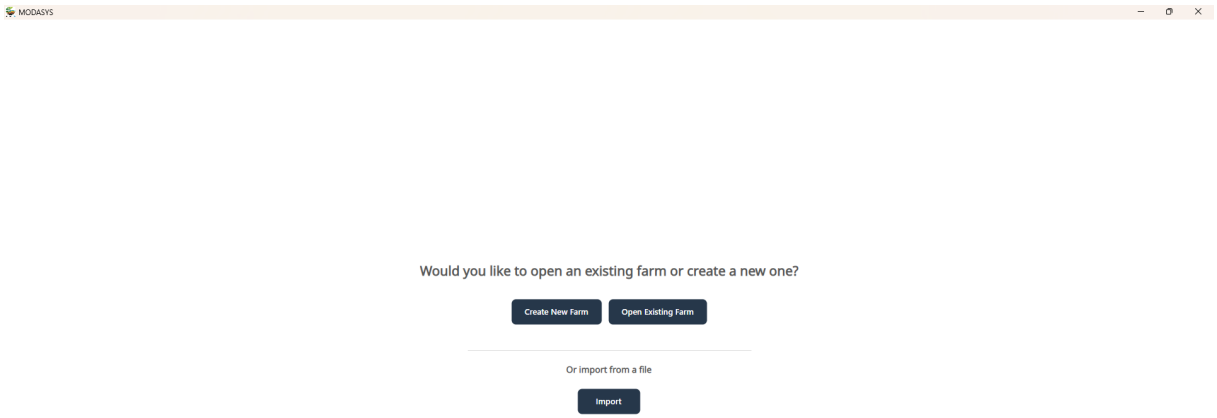



**Figure 2.2:** *The start screen:* ***Create New Farm****,* ***Open Existing Farm****, or* ***Import*** *a farm from a file.*

# 3. Creating or Opening a Farm

## 3.1 Creating a New Farm

1. Click **Create New Farm** on the start screen.
2. Enter a name for the farm and, optionally, comments, then click **Create New Farm** (Figure 3.1). The farm is saved straight away.
3. MODASYS then asks for the farm's climate region and country (Section 4).

New farms use metric units. The farm opens in **Basic** mode, which shows the essential inputs; Advanced mode can be switched on at any time (Section 5.3).

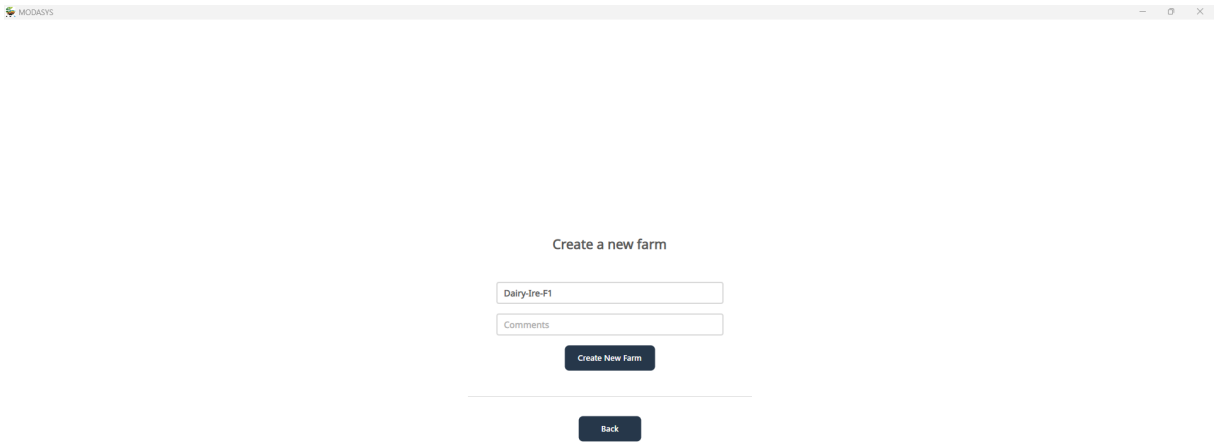



**Figure 3.1:** *Enter a name and comments for the new farm.*

## 3.2 Opening an Existing Farm

Click **Open Existing Farm** to see every farm stored on this computer, with the date it was created and last modified (Figure 3.2). Select a farm and click **Open**. **Remove** deletes the selected farm from the list.

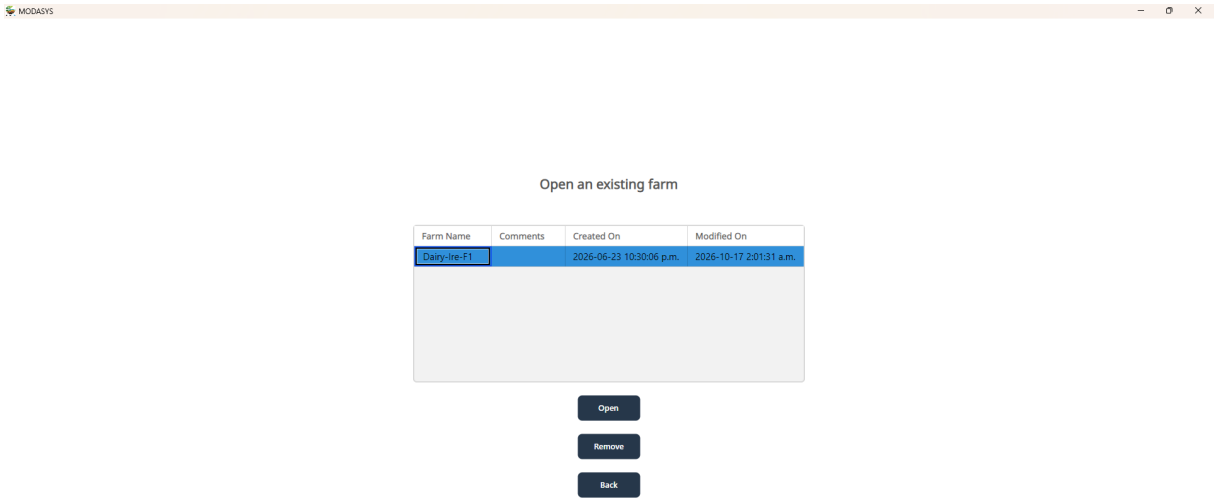



**Figure 3.2:** *Open an existing farm. Here the example farm Dairy-Ire-F1 is selected.*

### 3.3 Importing a Farm

Click **Import** on the start screen to load a farm that was exported from MODASYS (a `.modasys` file), for example one shared by a colleague. Farms can be exported from the Options page (**Export Farm**, Section 6), which also has its own **Import Farm** page.

## 4. Farm Location, Soil and Climate

### 4.1 Choosing the Climate Region and Country

MODASYS organises its data by climate region. The first location screen lists the 18 climate regions of the dataset roadmap (Step 1) and, for the selected region, the EU countries associated with it (Step 2). Countries whose data are already in MODASYS are marked **AVAILABLE NOW**; the others are marked **PLANNED**. At present Ireland is available (climate region *Cool Temperate Moist*). Select the region and country and click **Next** (Figure 4.1).

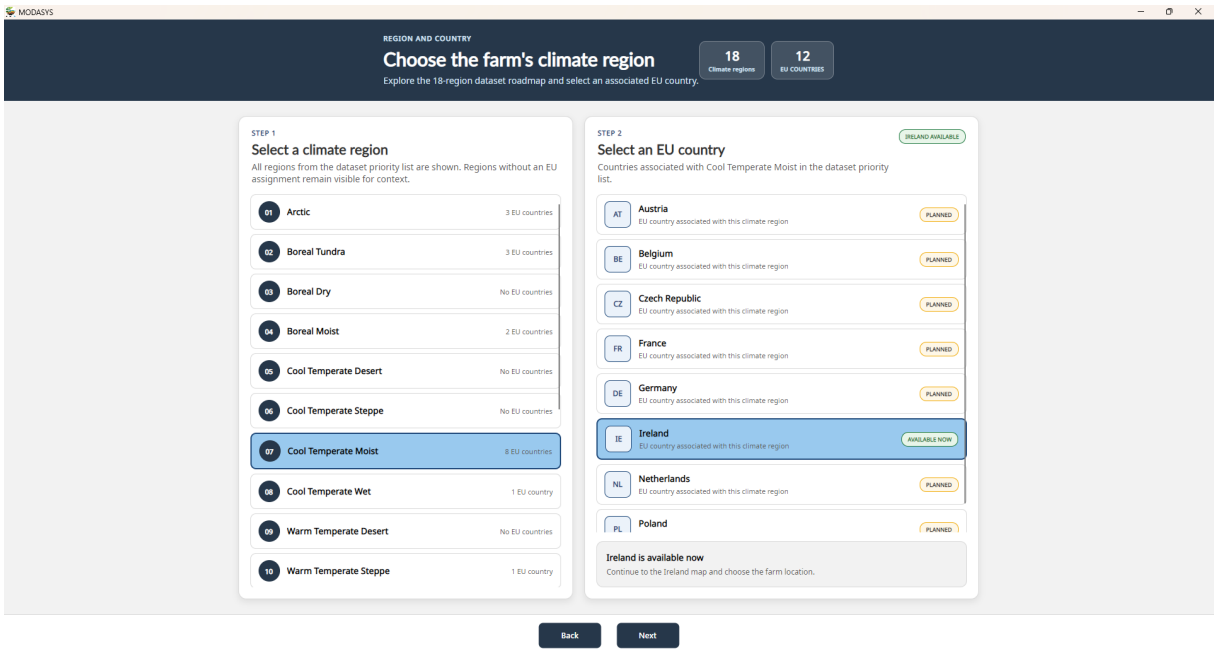



**Figure 4.1:** *Choose the farm's climate region (left) and EU country (right). Ireland is available now; other countries are planned.*

### 4.2 Using the Default Soil and Climate Data

The **Farm Location** screen shows a map with two tabs on the left, **Soil Data** and **Climate Data** (Figure 4.2).

1. On the **Soil Data** tab, choose the county from **Select County**. The soil polygons of that county are drawn in red on the map.
2. Zoom in with the mouse wheel (or the +/− buttons at the top right of the map) and **right-click** the polygon where the farm or land parcel is located.
3. MODASYS places a marker and loads the soil properties of that polygon from the integrated soil database (Figure 4.3): polygon ID, coordinates, ecozone, soil great group,

functional category, drainage class, texture, clay and sand content, soil organic carbon, pH, bulk density, top-layer thickness, organic carbon stock, total nitrogen and total phosphorus.

4. At the same time, daily climate data for the location are downloaded from the **NASA POWER** daily API. The **Climate Data** tab shows the data source, coordinates and the period covered (Figure 4.4).
5. Click **Next**. MODASYS opens the Component Selection screen of the new farm.

**Clear** (top right) removes the selected location so you can pick again.

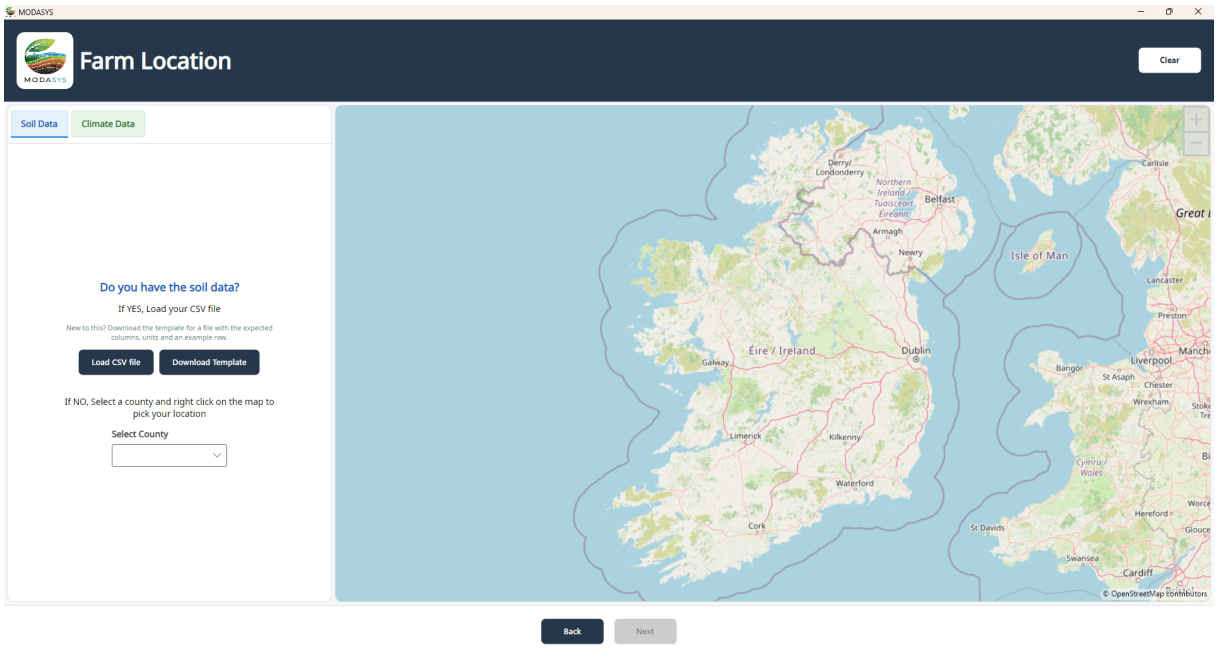



**Figure 4.2:** *The Farm Location screen before a location is chosen. Load your own soil file, or select a county and right-click the map.*

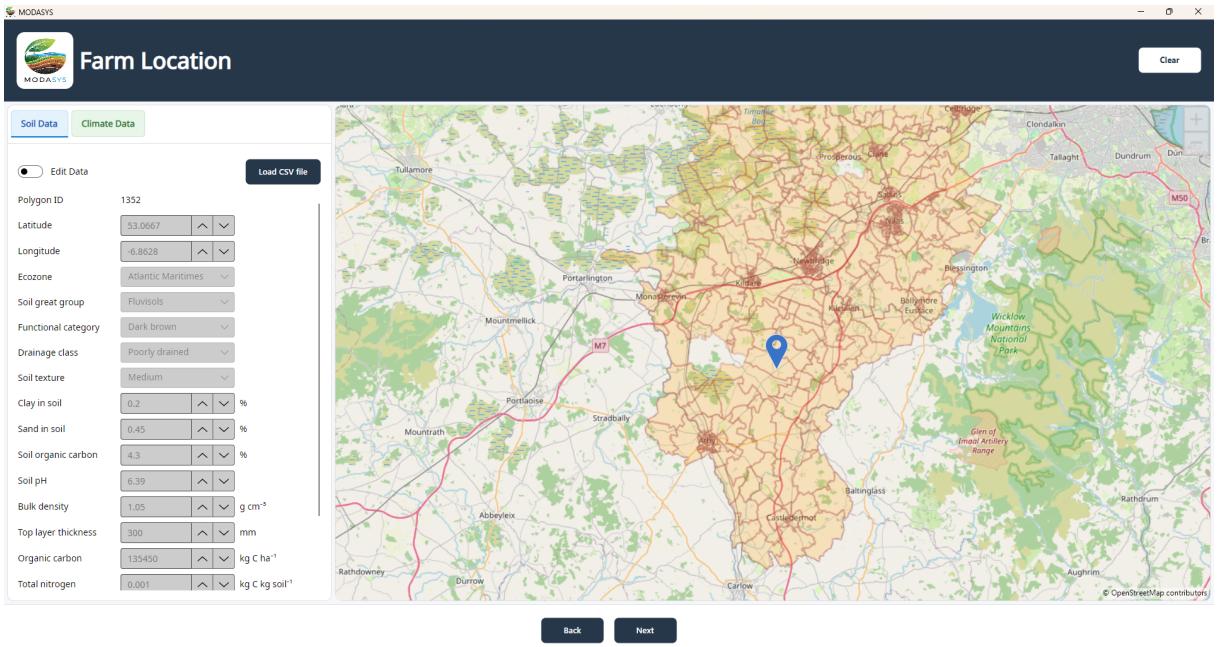



**Figure 4.3:** *Polygon 1352 in Co. Kildare, the location of Dairy-Ire-F1, has been right-clicked. The marker shows the location and the soil properties of the polygon appear on the left.*

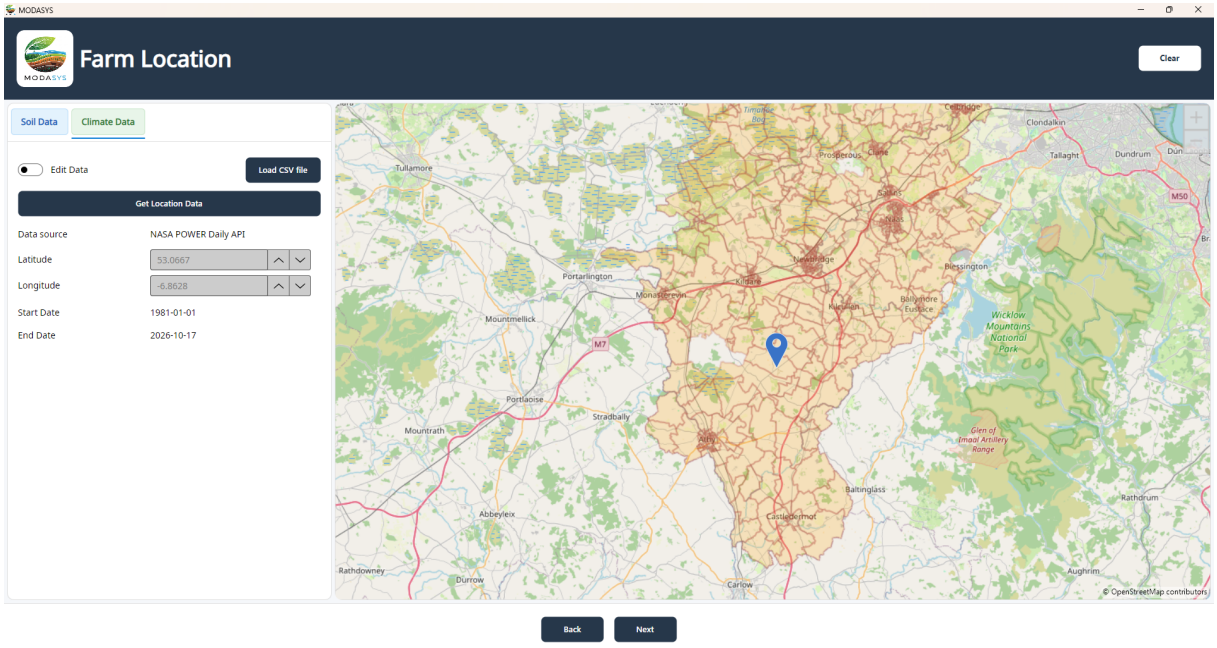



**Figure 4.4:** *The Climate Data tab after a location is chosen: daily climate from the NASA POWER API for the selected coordinates.*

## 4.3 Using Your Own Soil Data

Users with measured soil data, and users outside the countries already available, should enter their own soil properties so the model represents their site.

- **From a file:** on the **Soil Data** tab click **Download Template** to save a CSV file with

the expected columns, units and an example row. Fill in your values and click **Load CSV file**.

- **By hand:** after a location is chosen, switch on **Edit Data** and type over the default values.
- **For one field only:** open the field's **Soil** tab in the component editor (Section 5.4), switch on **Edit Data** and change the properties for that field.

## 4.4 Using Your Own Climate Data

On the **Climate Data** tab (Figure 4.5) you can either fetch the default NASA POWER data for the coordinates (**Get Location Data**) or click **Load CSV file** to use your own daily climate record. MODASYS confirms when the file has loaded.

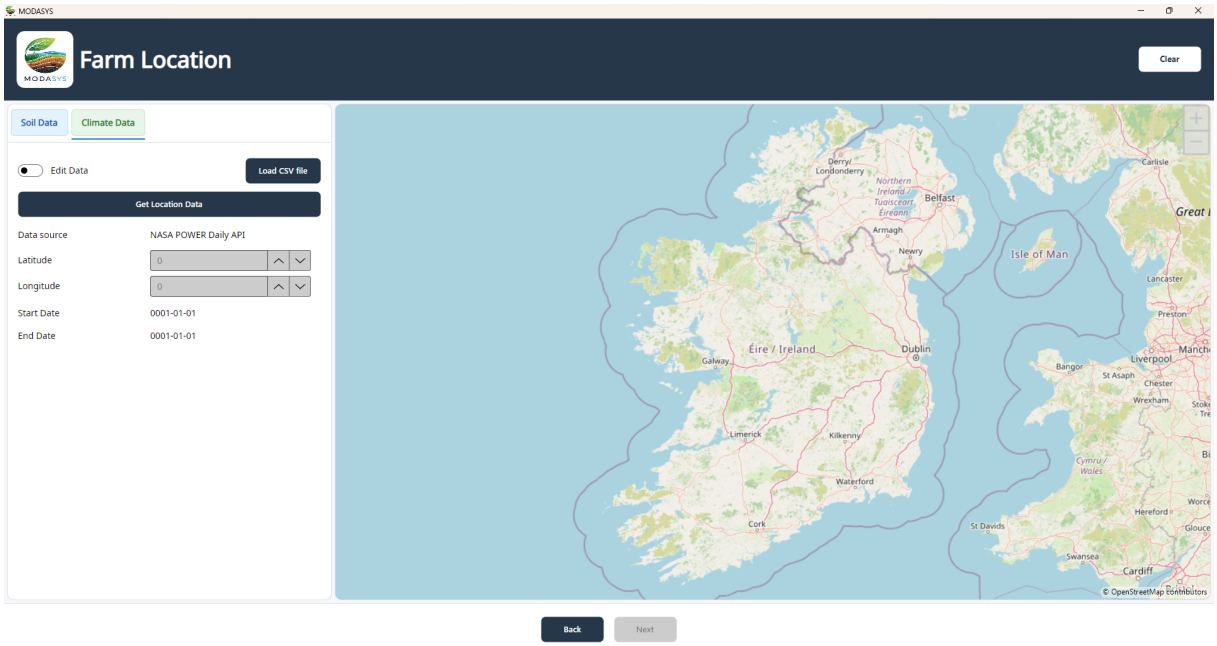



**Figure 4.5:** *The Climate Data tab, where you can load your own daily climate file.*

The climate file must be a comma-separated values file with the `.csv` extension. The first row must contain exactly these five column headings, in this order (capitals and spaces are ignored), and each following row holds one day:

**Table 4.1:** *Columns of a custom climate data file.*

| Heading | Content |
|---|---|
| Year | The calendar year of the record. |
| Julian Day | Day of the year (1–365, or 366 in a leap year). |
| Mean Daily Air Temperature | Mean air temperature of the day (°C). |
| Mean Daily Precipitation | Precipitation of the day (mm). |
| Mean Daily PET | Potential evapotranspiration of the day (mm). |

For example:

```
Year,Julian Day,Mean Daily Air Temperature,Mean Daily Precipitation,Mean Daily PET
2020,1,6.2,3.4,0.5
2020,2,5.8,0.0,0.6
```

# 5. Building the Farm: Components

## 5.1 The Component Selection Screen

The farm is described by its components. The Component Selection screen (Figure 5.1) has three parts:

- **Available Components** (left column): every component type, grouped into **Site Characteristics** (Field), **Land Use and Management** (Crop, Grass, Agroforestry), **Livestock** (beef and dairy cattle systems such as Store Cattle System, Suckler Beef Production, Beef Finisher and Dairy Farming; pig systems such as Piglet Production, Isowean, Farrow-to-Finish and Grower-Finisher; scroll down for the other livestock types) and **Infrastructure** (Farm Infrastructure, including anaerobic digestion).
- **My Components** (middle column): the components added to this farm, with their count on the **My Components** button. Click a component to open it; click × to remove it.
- **The component editor** (right): the inputs of the selected component.

The buttons along the top are **SAVE** (save the farm), the **BASIC/ADVANCED** switch, and the gear icon, which opens the Options page. The panel icon next to **My Components** hides the two component columns to give the editor the full width. **Back** returns to the location screen and **Results** runs the model and opens the results.

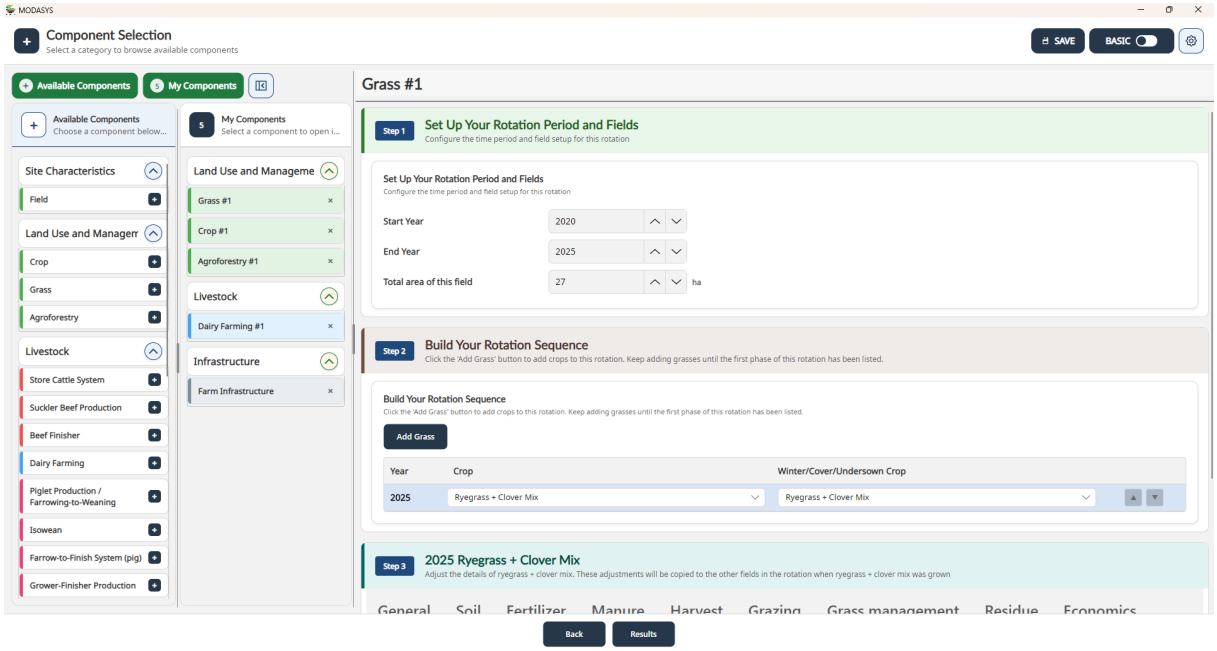



**Figure 5.1:** *The Component Selection screen for Dairy-Ire-F1: available components (left), the farm's five components (middle) and the editor for the selected component, Grass #1 (right).*

## 5.2 Adding a Component

Click the + next to a component in **Available Components**. The new component appears under **My Components** with a number (e.g. Field #1) and opens in the editor with default values (Figure 5.3). Add as many components as the farm needs: several fields, a mix of crops and grass, one or more herds, agroforestry and infrastructure. The model is run once for the whole farm, and the results can be read for single components or for the mixed system.

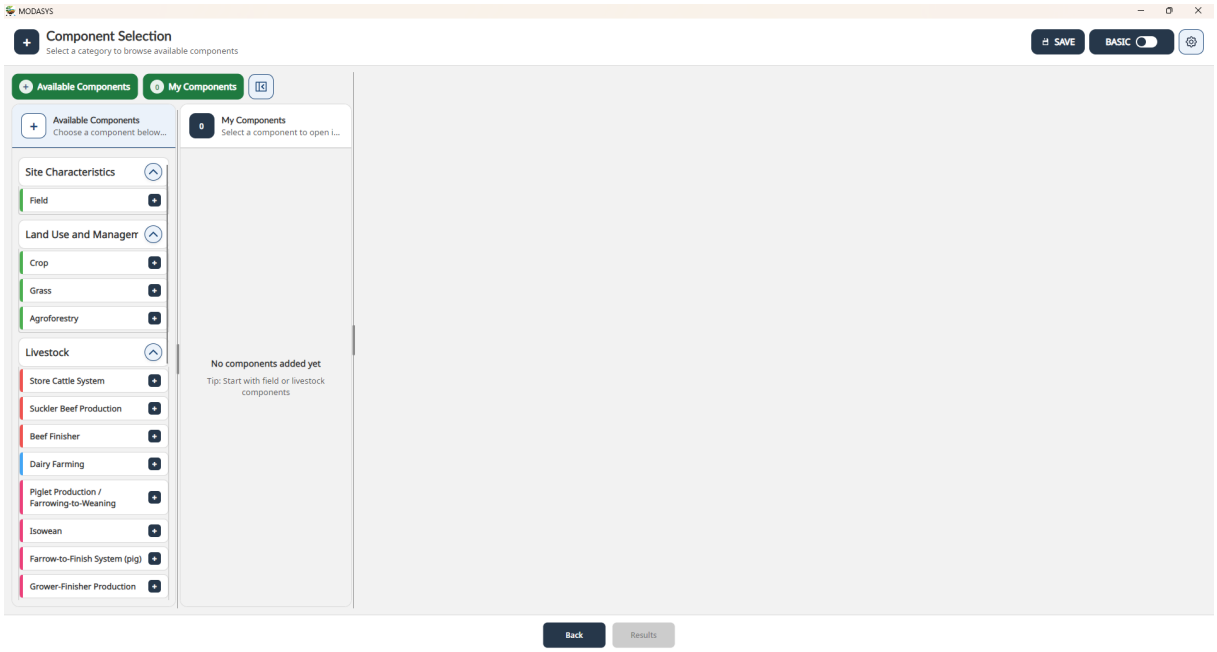



**Figure 5.2:** *A new farm before any component has been added.*

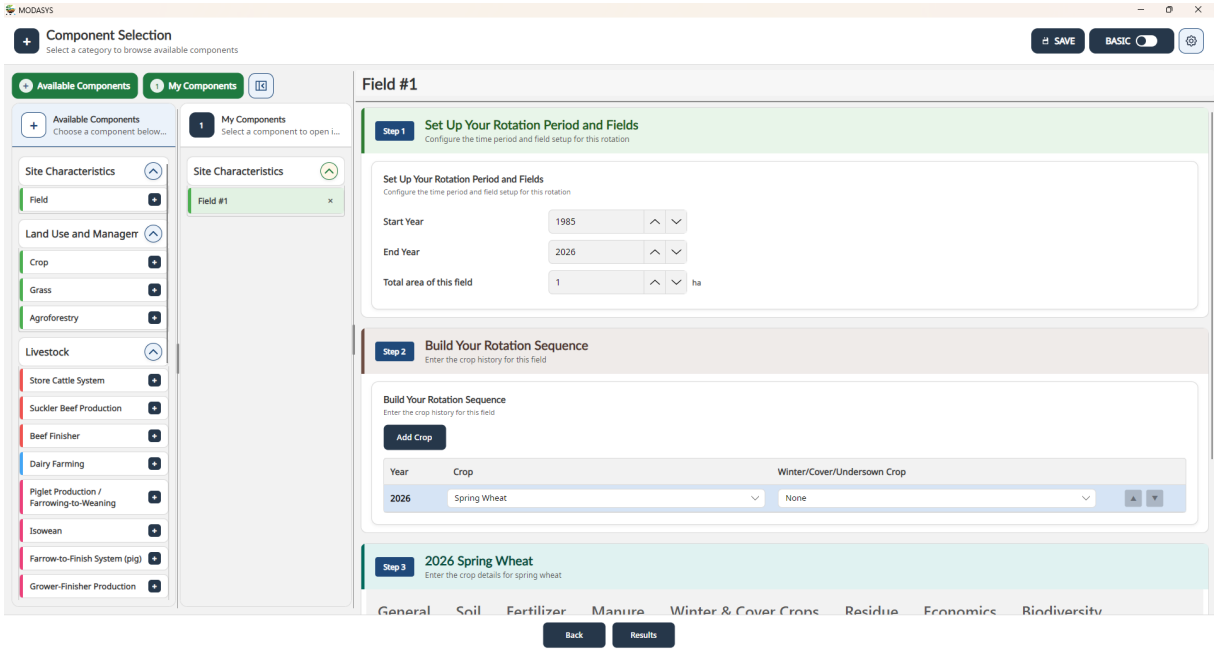



**Figure 5.3:** *After clicking + next to Field, Field #1 appears under My Components and opens in the editor.*

## 5.3 Basic and Advanced Mode

Farms open in **Basic** mode, which shows the inputs most users need. Click the **BASIC** switch at the top right to change to **ADVANCED** mode, which adds further inputs such as the fuel and herbicide energy of a crop (Figure 5.4). The mode is saved with the farm, and the **Operation Mode** toggle in **Farm Settings** is the same setting.

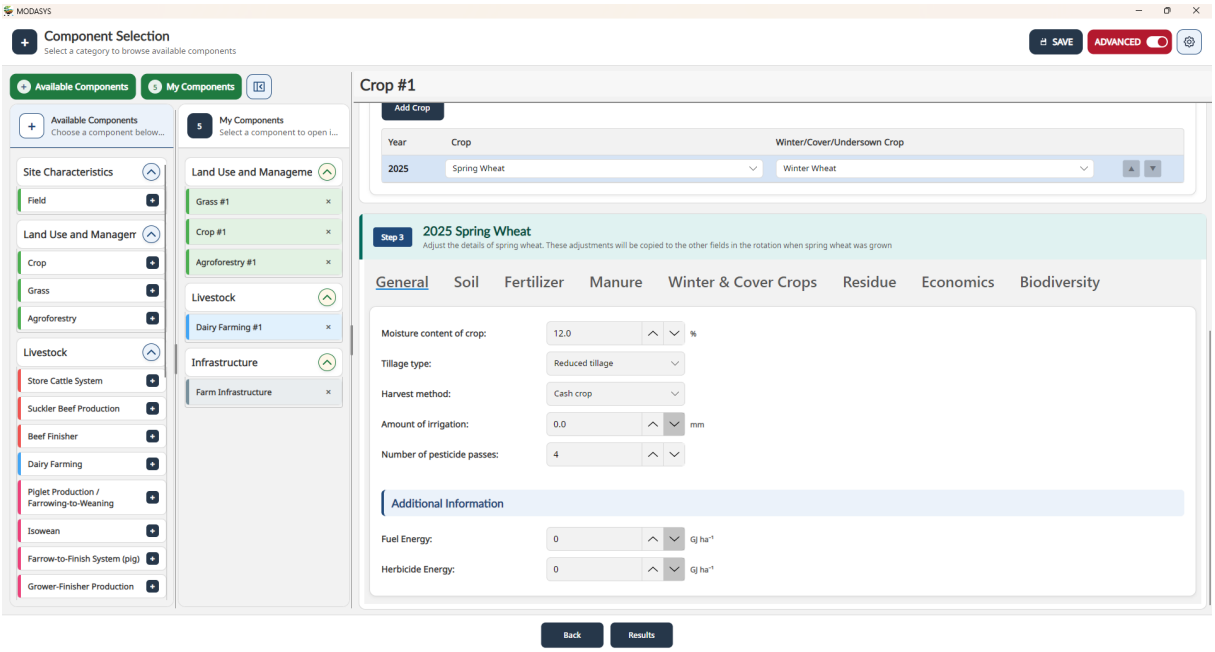



**Figure 5.4:** *Advanced mode (red switch, top right) adds the* Additional Information *section to a crop's General tab.*

## 5.4 Fields: Crop and Grass Components

Field, Crop and Grass components share the same three-step layout (Figure 5.5):

**Step 1 – Set up your rotation period and fields.** The start and end year of the rotation and the total area of the field (ha).

**Step 2 – Build your rotation sequence.** Click **Add Crop** (or **Add Grass**) to add a year to the rotation, then choose the main crop and any winter, cover or undersown crop for that year. The arrows reorder the years.

**Step 3 – Crop details.** The details of the crop selected in Step 2, in a set of tabs. Changes are copied to the other years of the rotation in which the same crop was grown.

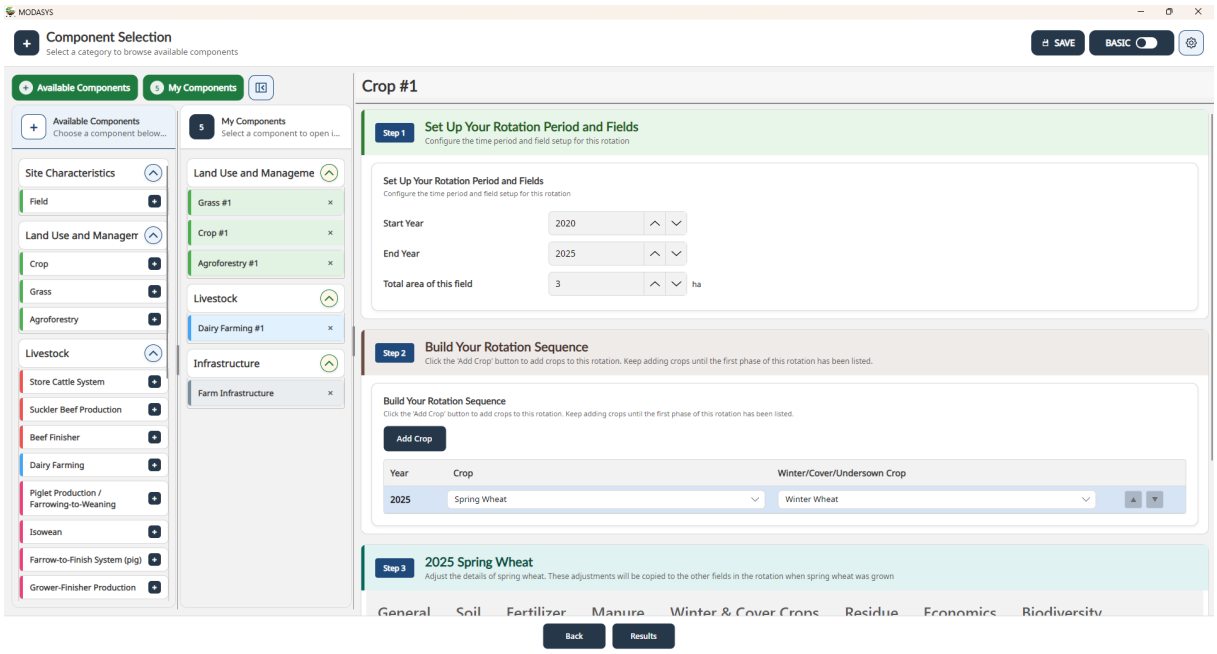



**Figure 5.5:** *Steps 1 and 2 of a crop component: the rotation period, field area and rotation sequence.*

## Crop tabs

**General** Moisture content of the crop, tillage type, harvest method, amount of irrigation and number of pesticide passes (Figure 5.6).

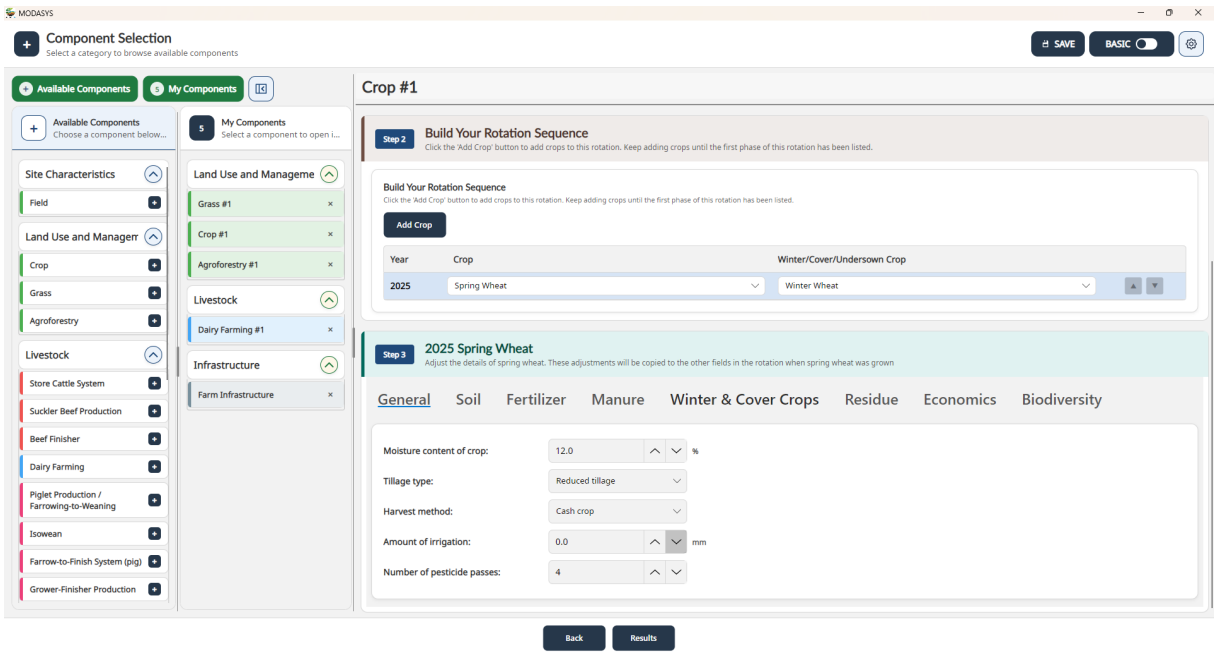



**Figure 5.6:** *General tab of a crop component.*

**Soil** The soil of this field, taken from the farm location. Switch on **Edit Data** to change the soil properties for this field only (Figure 5.7).

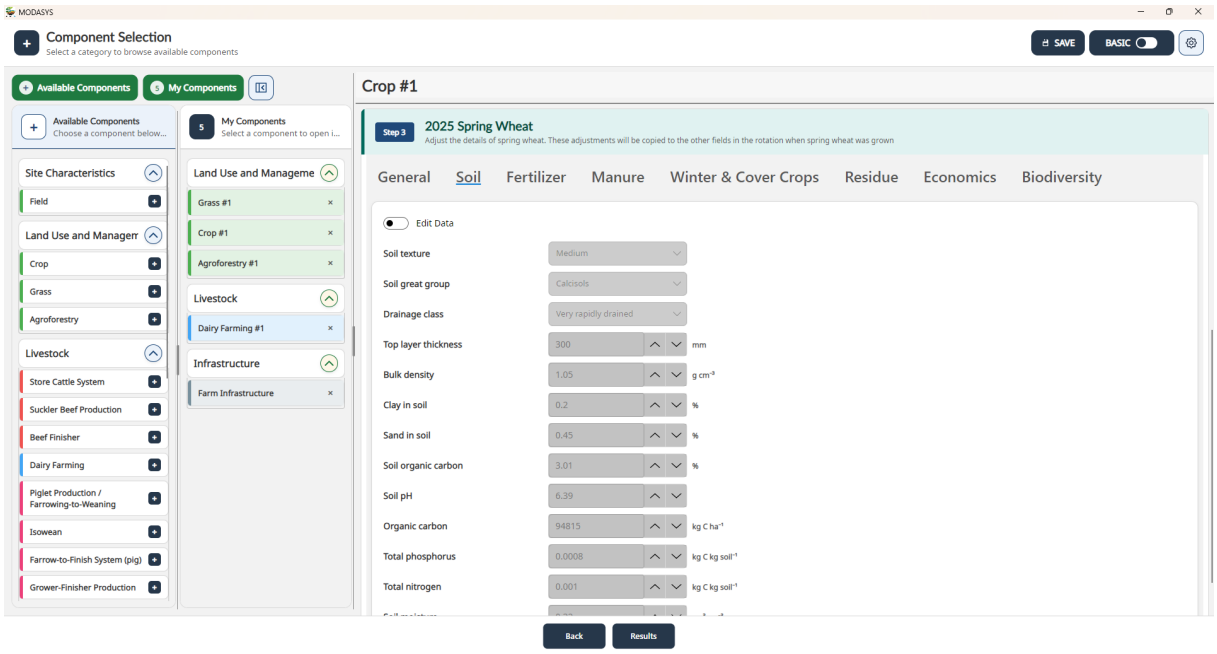



**Figure 5.7:** *Soil tab of a crop component.*

**Fertilizer** One card per fertilizer application: date, fertilizer blend, application method and rate (kg $ha^{-1}$). Applications can be added and removed (Figure 5.8).

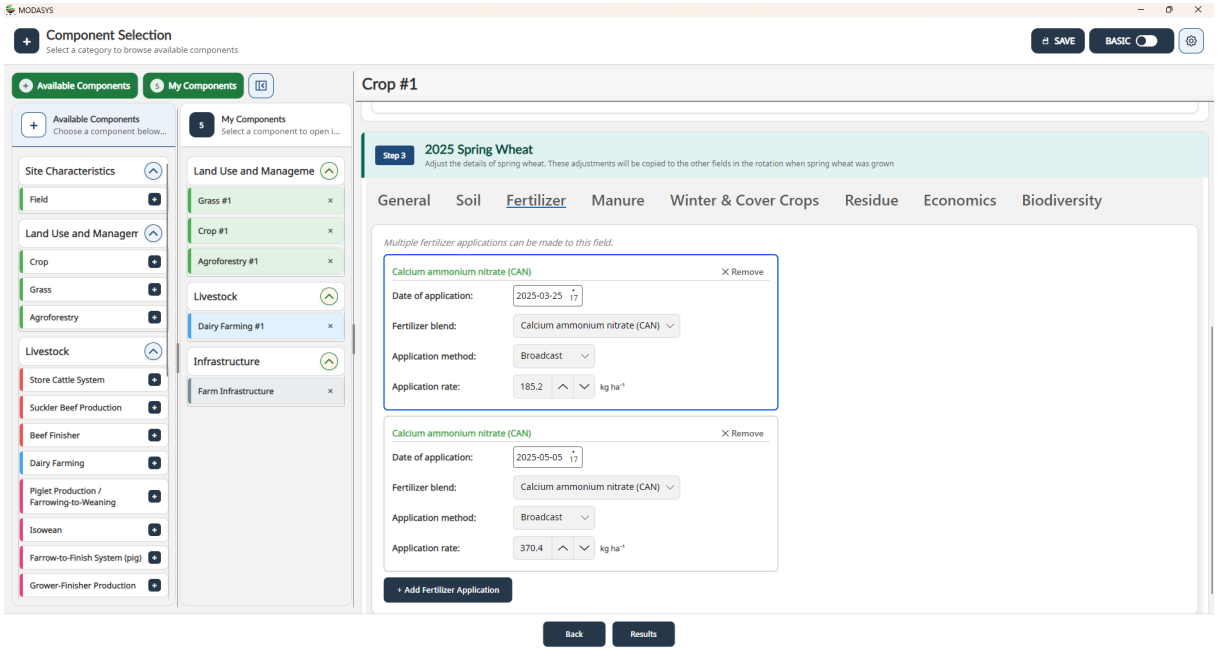



**Figure 5.8:** *Fertilizer tab of a crop component.*

**Manure** One card per manure application: date, origin of the manure (livestock on this farm or imported), manure type, manure handling system, application method, and the amount of manure and nitrogen applied (Figure 5.9).

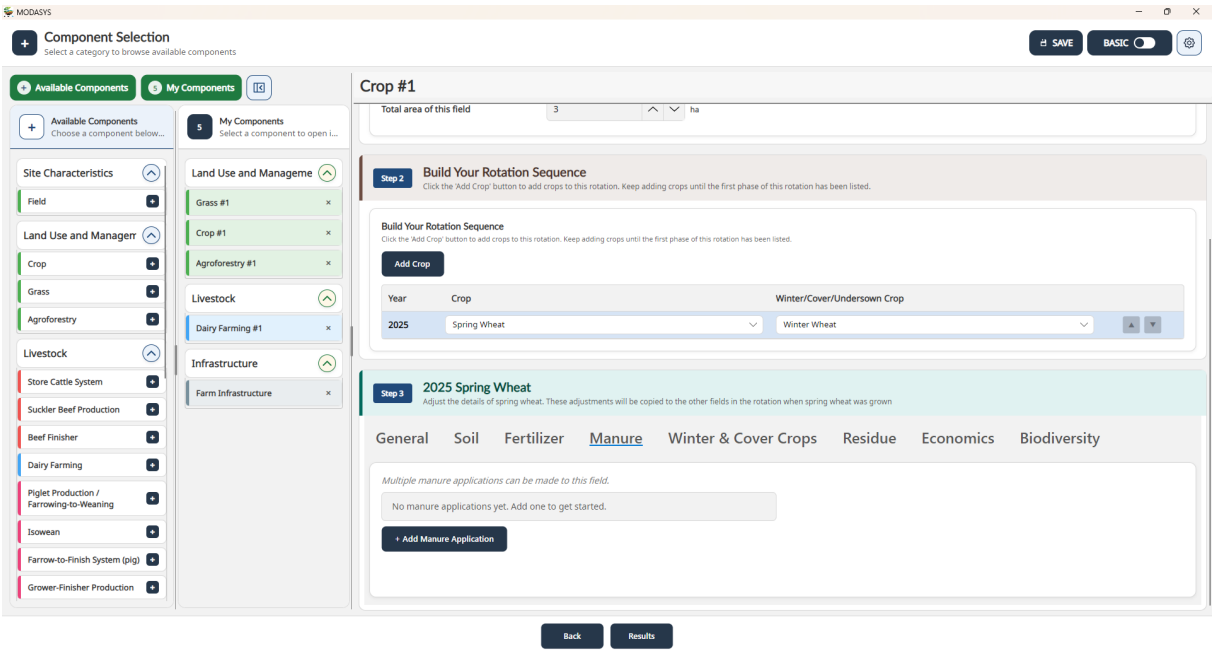



**Figure 5.9:** *Manure tab of a crop component.*

**Winter & Cover Crops** Details of any winter, cover or undersown crop grown in the field (Figure 5.10).

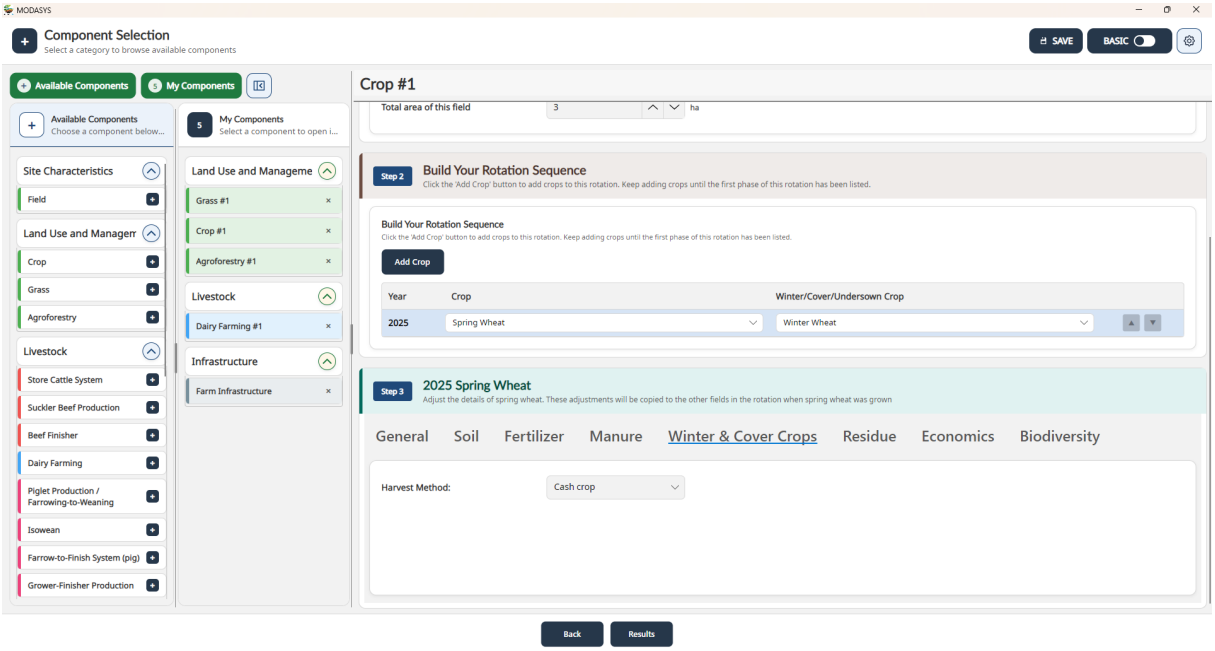



**Figure 5.10:** *Winter & Cover Crops tab.*

**Residue** Biomass coefficients (product, straw, roots, extra-roots), the percentage of each returned to the soil, and their nitrogen contents. These drive the soil carbon and nitrogen inputs (Figure 5.11).

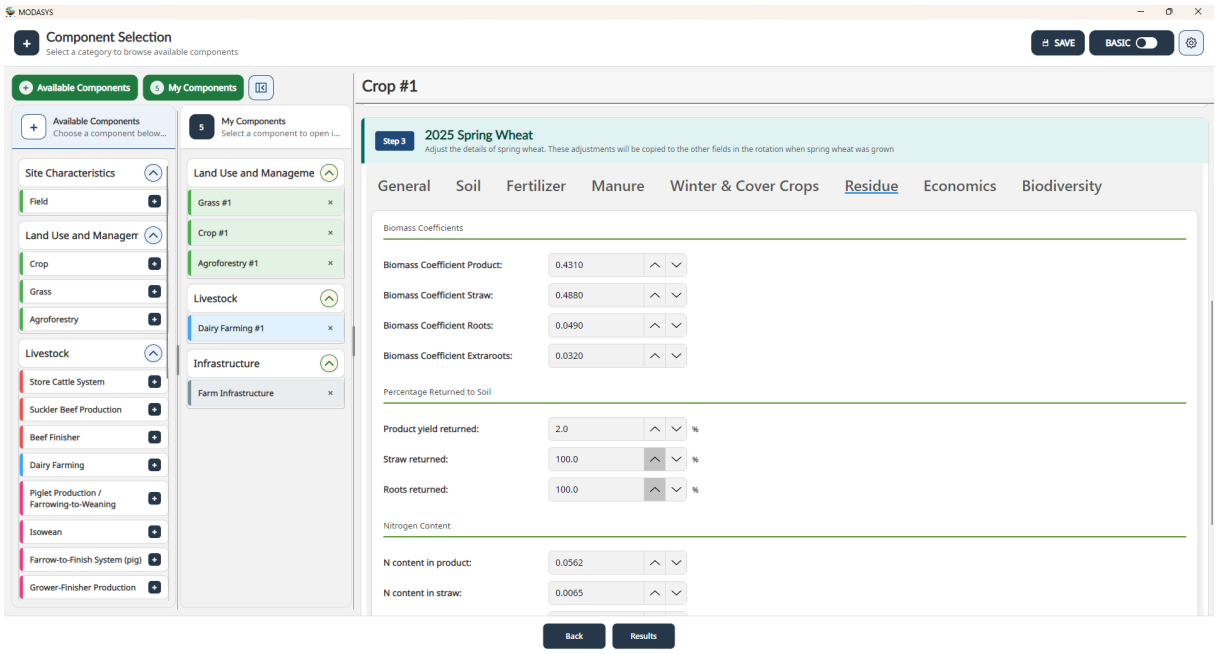



**Figure 5.11:** *Residue tab of a crop component.*

**Economics** Direct costs per hectare: seed, fertilizer, chemicals, trucking and marketing, fuel, machinery and building repairs, insurance, custom work, labour, utilities and operating interest (Figure 5.12).

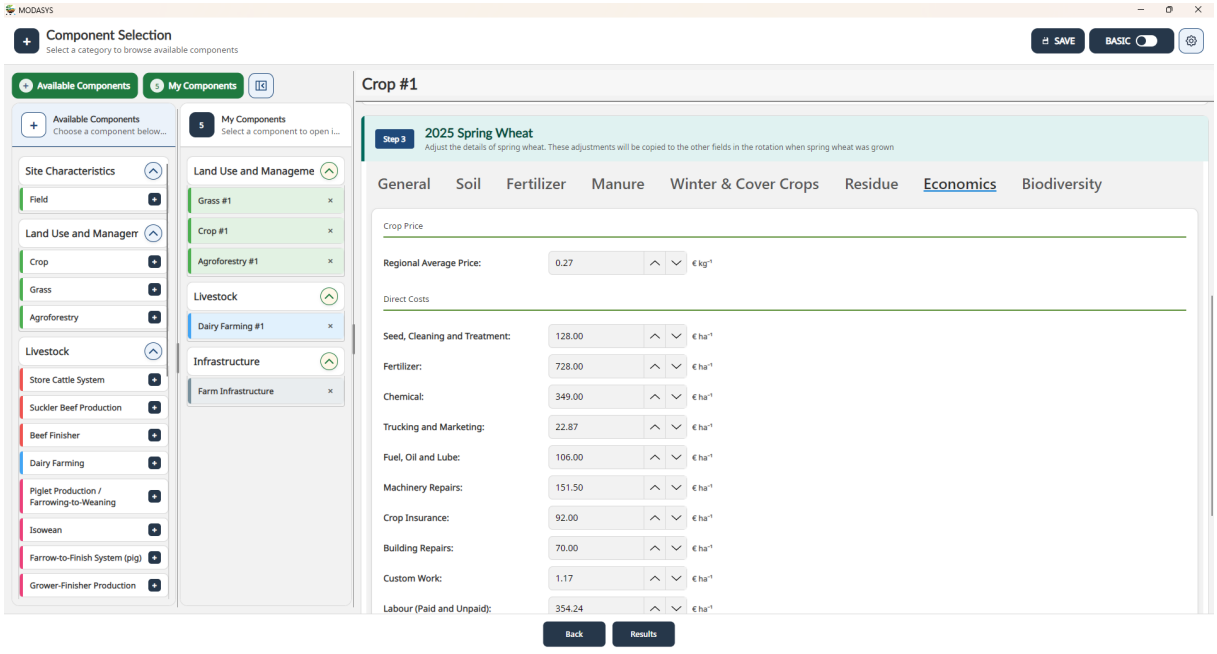



**Figure 5.12:** *Economics tab of a crop component.*

**Biodiversity** Inputs for the biodiversity index: pesticide ecotoxicity (CTUe $ha^{-1}$ $yr^{-1}$) and diesel used on the field (Figure 5.13). Leave diesel at zero to use the farm-level diesel figure instead; the two are never added together.

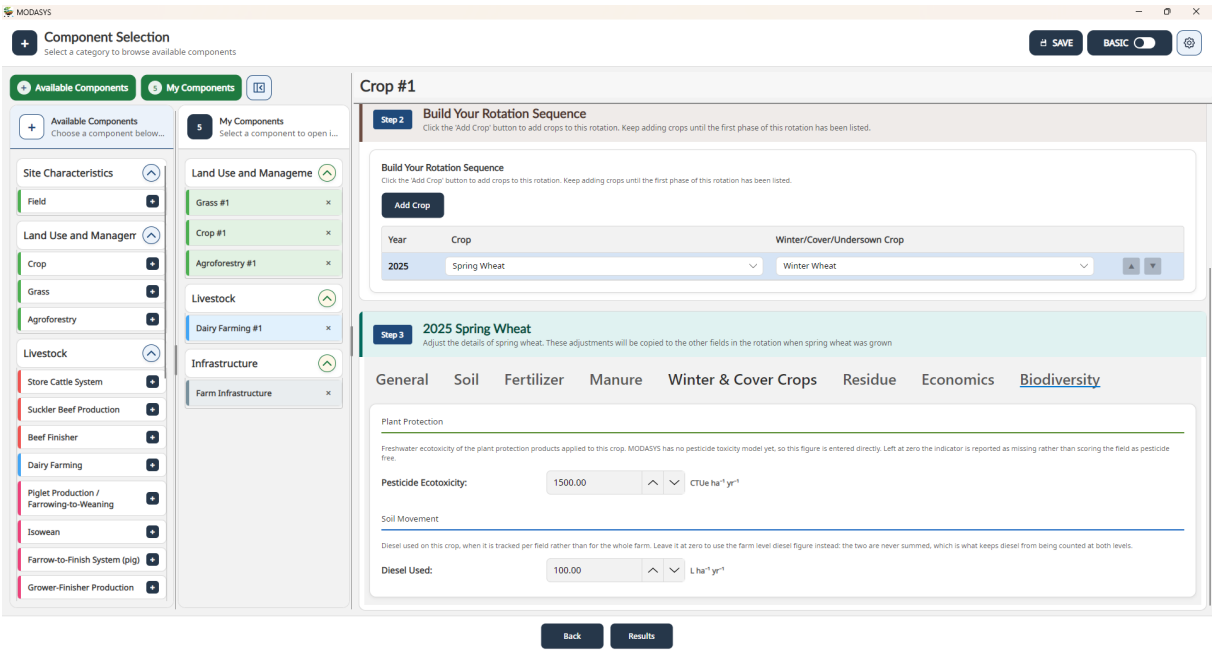



**Figure 5.13:** *Biodiversity tab of a crop component.*

### Additional grass tabs

A grass component has the same tabs as a crop, plus three for grassland management (Figures 5.14–5.23).

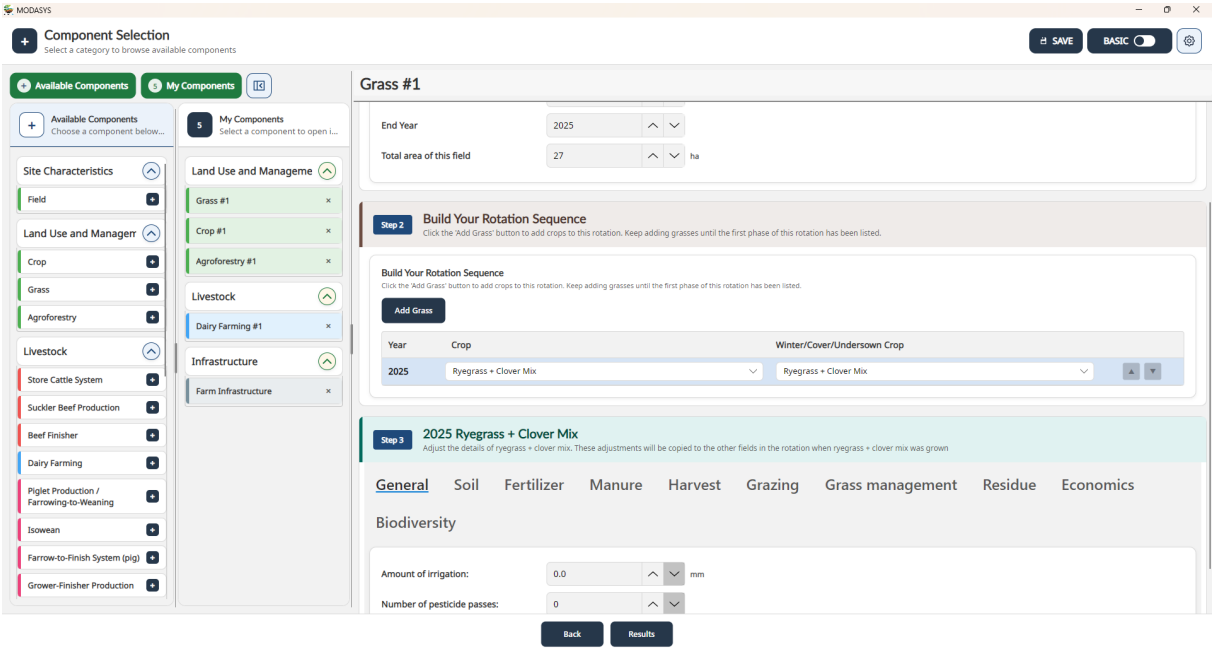



**Figure 5.14:** *Grass component: rotation sequence, Step 3 and the General tab.*

**Harvest** One card per cut for silage or hay: date, forage growth stage, number of bales, wet bale weight, harvest loss and moisture content. The total harvested biomass is shown on the right (Figure 5.15).

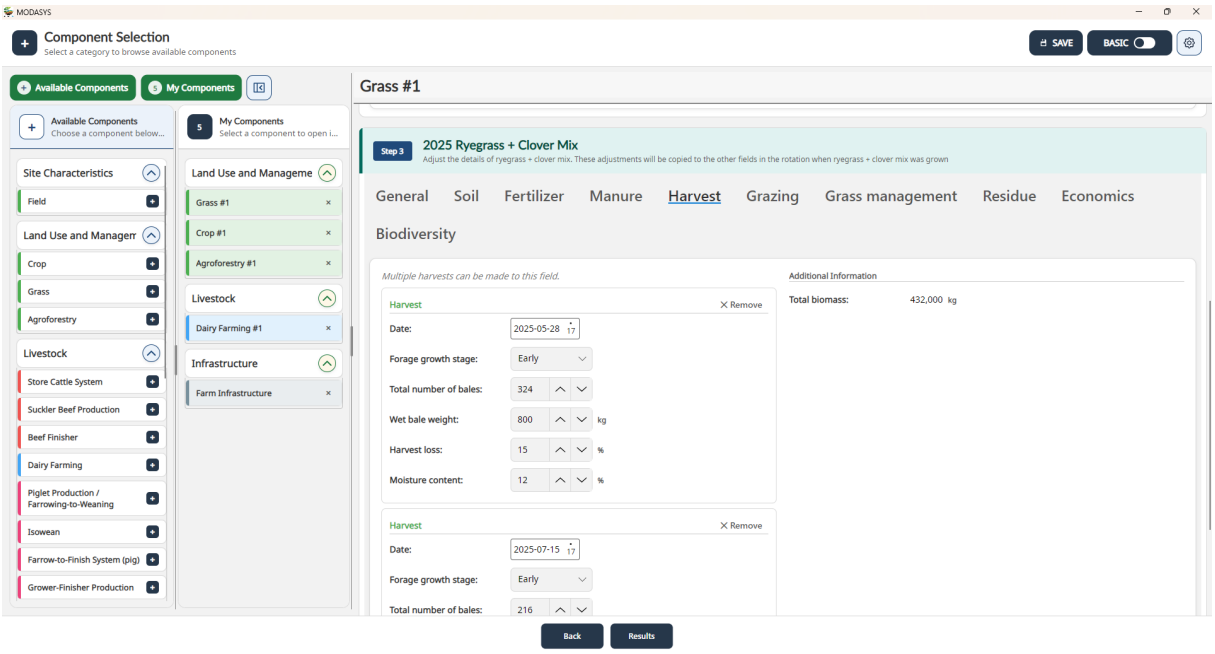



**Figure 5.15:** *Harvest tab of a grass component.*

**Grazing** The animal groups that graze this field. This list is derived from the livestock components: to change a grazing period, edit the animal group's management period and its Housing tab (Section 5.6) (Figure 5.16).

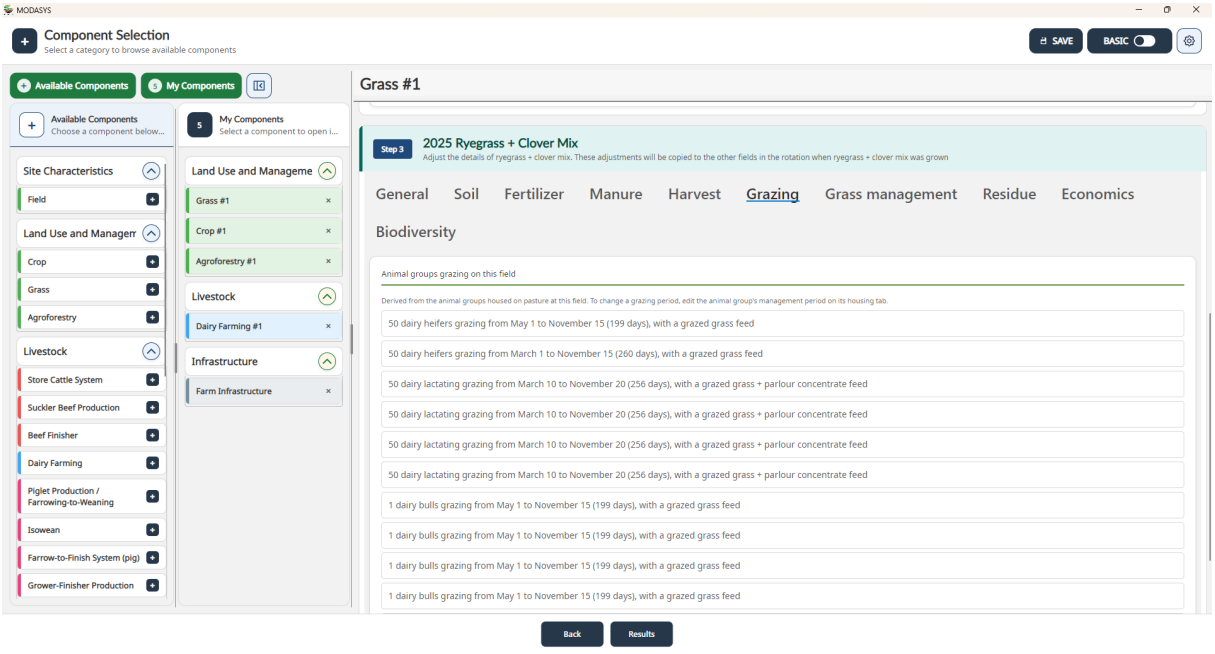



**Figure 5.16:** *Grazing tab: the dairy groups grazing this field, with their periods and feeds.*

**Grass management** Inputs for the grass growth model: the management type (cutting, mowing, grazing, or mixed cutting and grazing), the starting biomass, and up to four cuts, mowings or grazings per season with their day of year and cutting height (Figure 5.17). If none are entered, the model uses the field's harvest and grazing records.

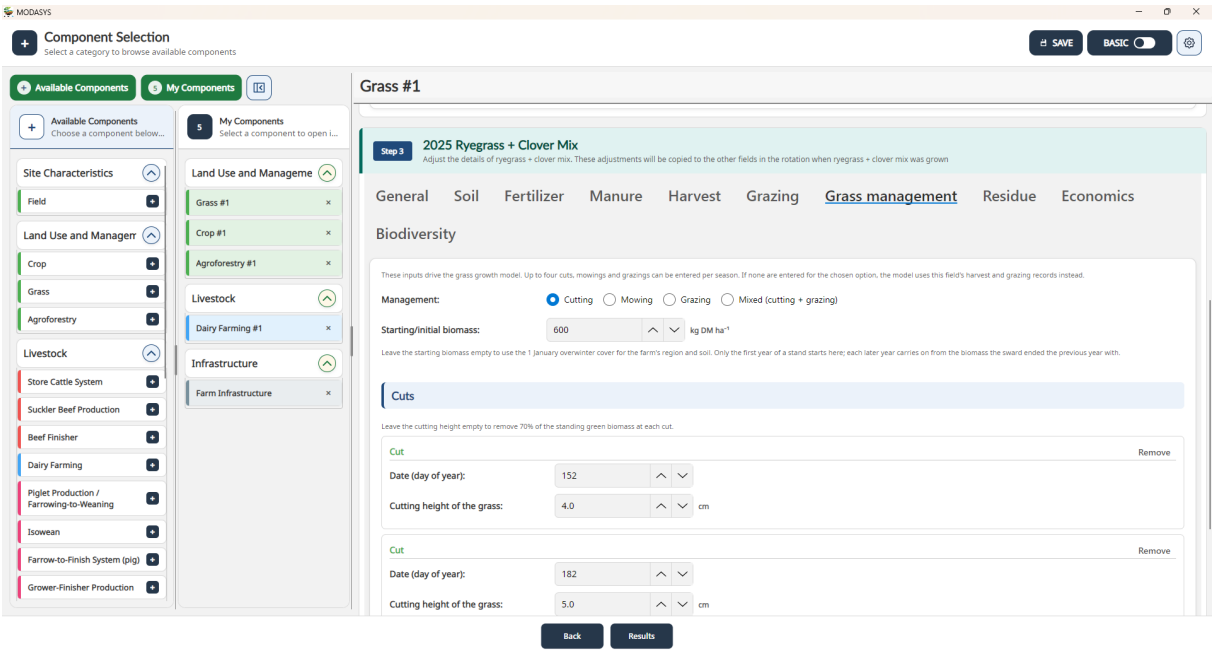



**Figure 5.17:** *Grass management tab, which drives the grass growth model.*

The Soil, Fertilizer, Manure, Residue, Economics and Biodiversity tabs work as for crops:

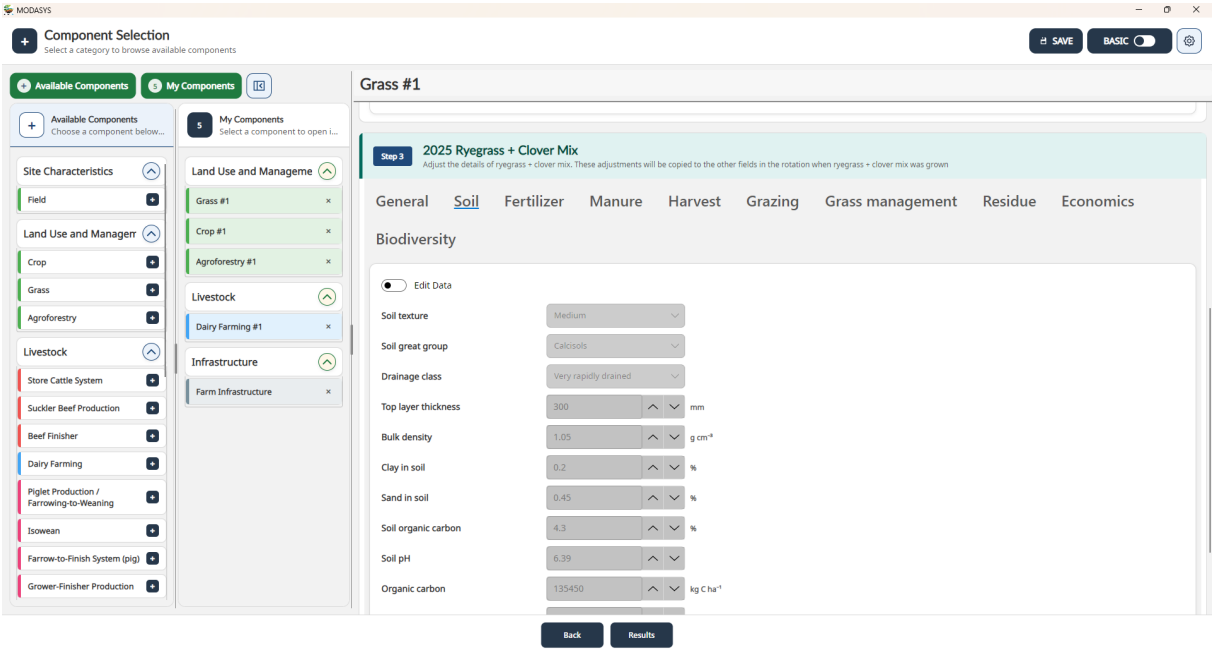



**Figure 5.18:** *Soil tab of a grass component. Switch on Edit Data to change this field's soil.*

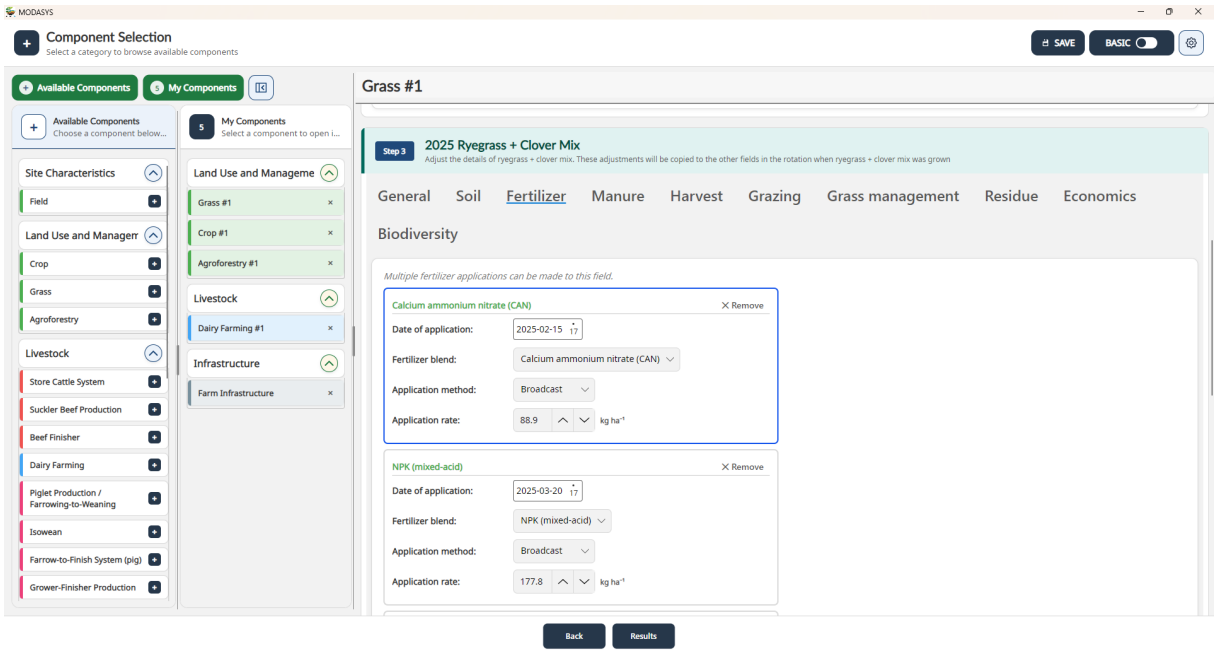



**Figure 5.19:** *Fertilizer tab of a grass component, with several applications through the season.*

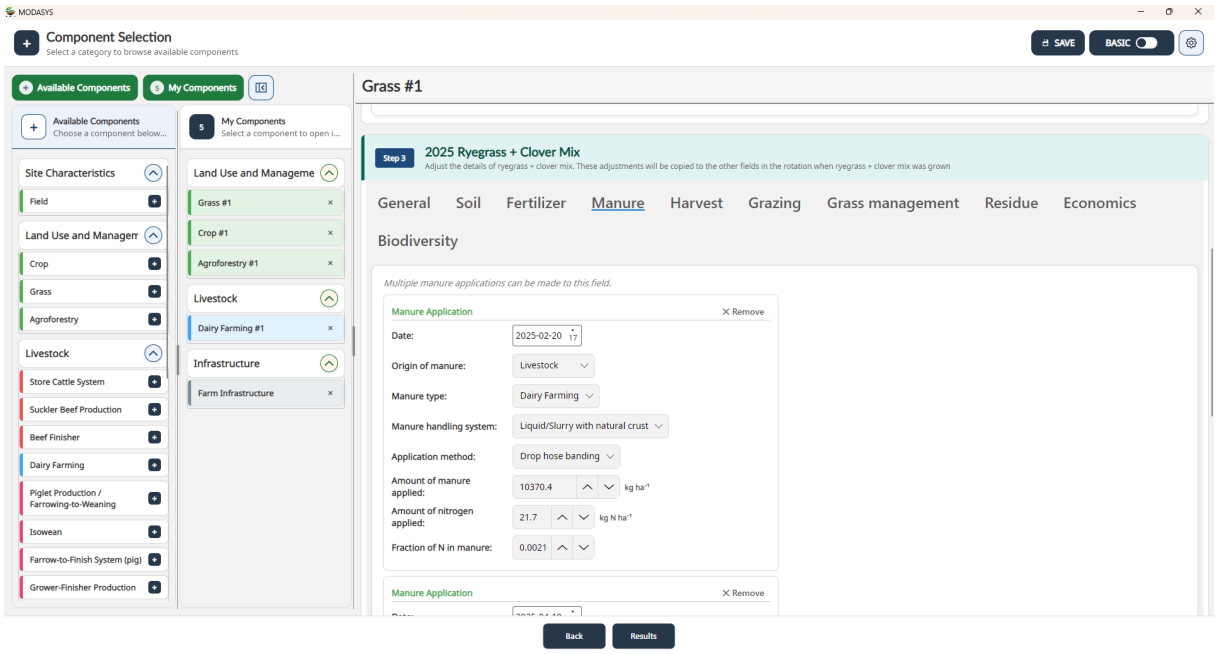



**Figure 5.20:** *Manure tab of a grass component: dairy slurry applied by drop-hose banding in February (10.4 t ha$^{-1}$), before closing for silage in April and after first-cut silage in June (5.6 t ha$^{-1}$ each). The N applied follows from the N fraction of the manure.*

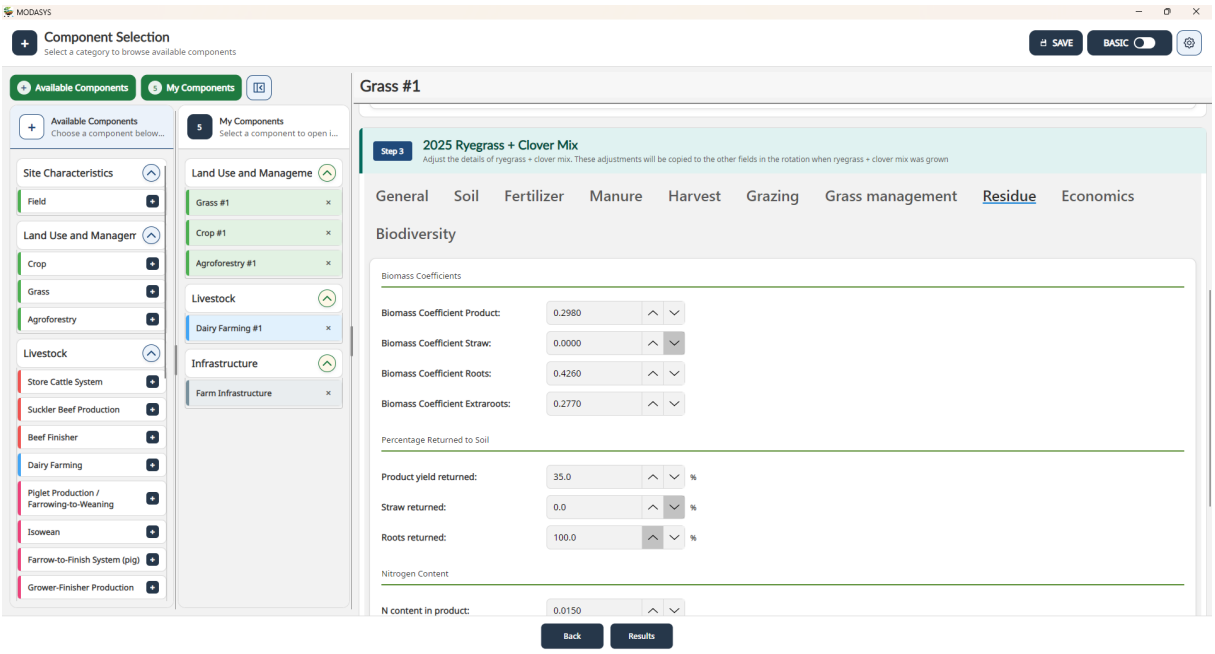



**Figure 5.21:** *Residue tab of a grass component.*

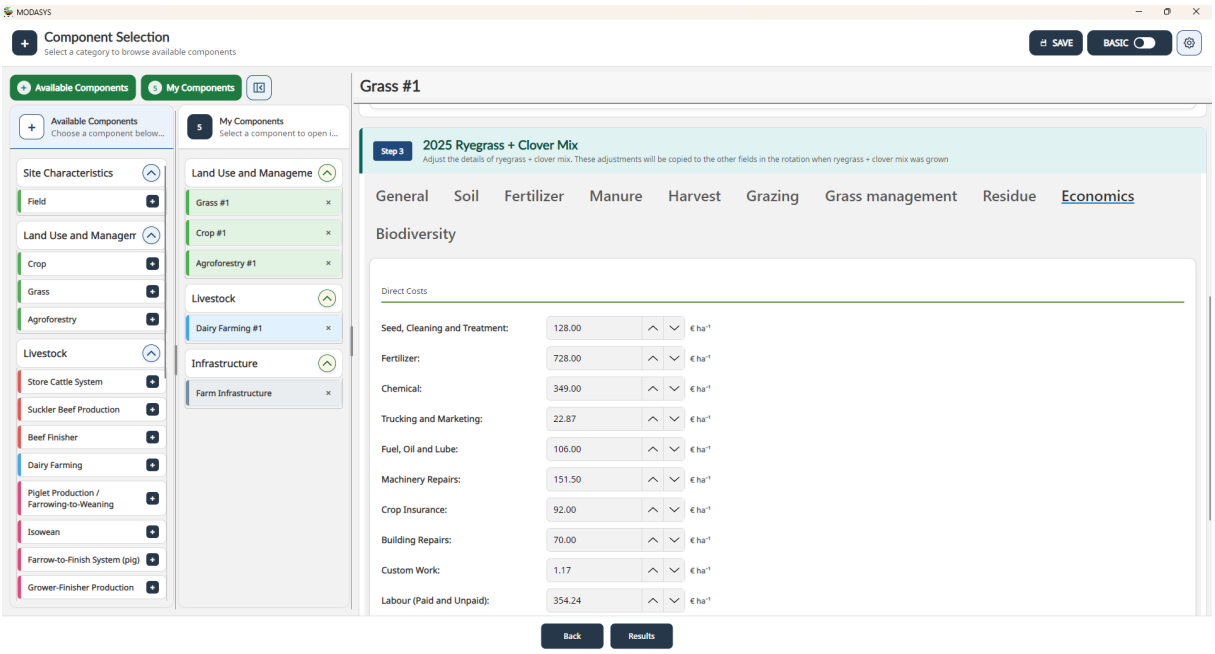



**Figure 5.22:** *Economics tab of a grass component.*

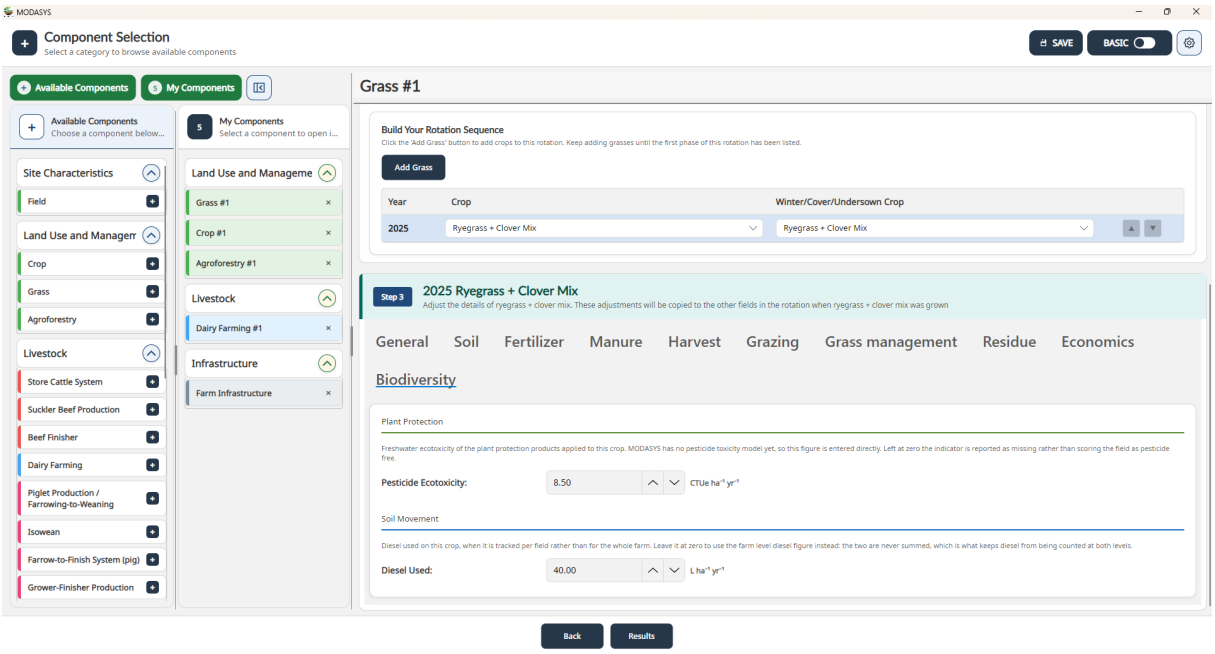



**Figure 5.23:** *Biodiversity tab of a grass component.*

## 5.5 Agroforestry Component

The agroforestry component models trees on the farm as a carbon sequestration measure (Figure 5.24).

**Step 1 – Agroforestry system type.** Silvoarable, silvopastoral, agrosilvopastoral, hedgerows, orchard or farm garden.

**Step 2 – Tree and plantation details.** Observation year, tree species, planting year, seedling age, average tree height, diameter at breast height, number of trees and cutting year.

**Step 3 – Plantation layout, management and economics.** A set of tabs, described below.

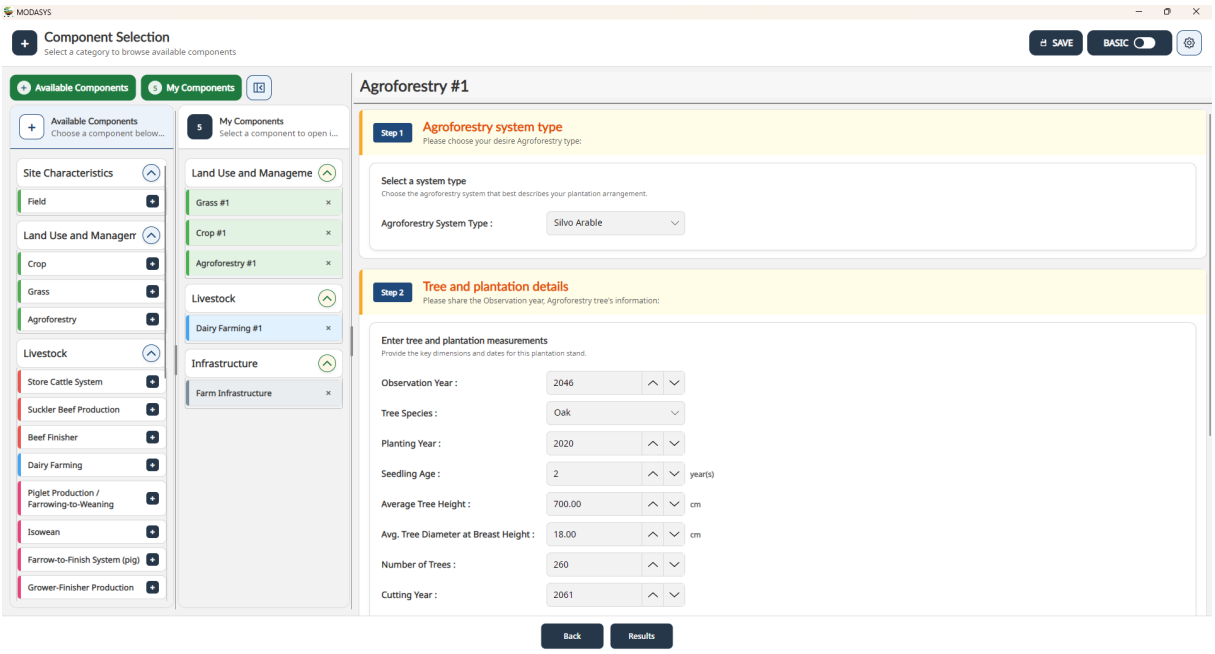



**Figure 5.24:** *Steps 1 and 2 of the agroforestry component.*

**Plantation Area** The plantation type (e.g. circular) and its dimensions (Figure 5.25).

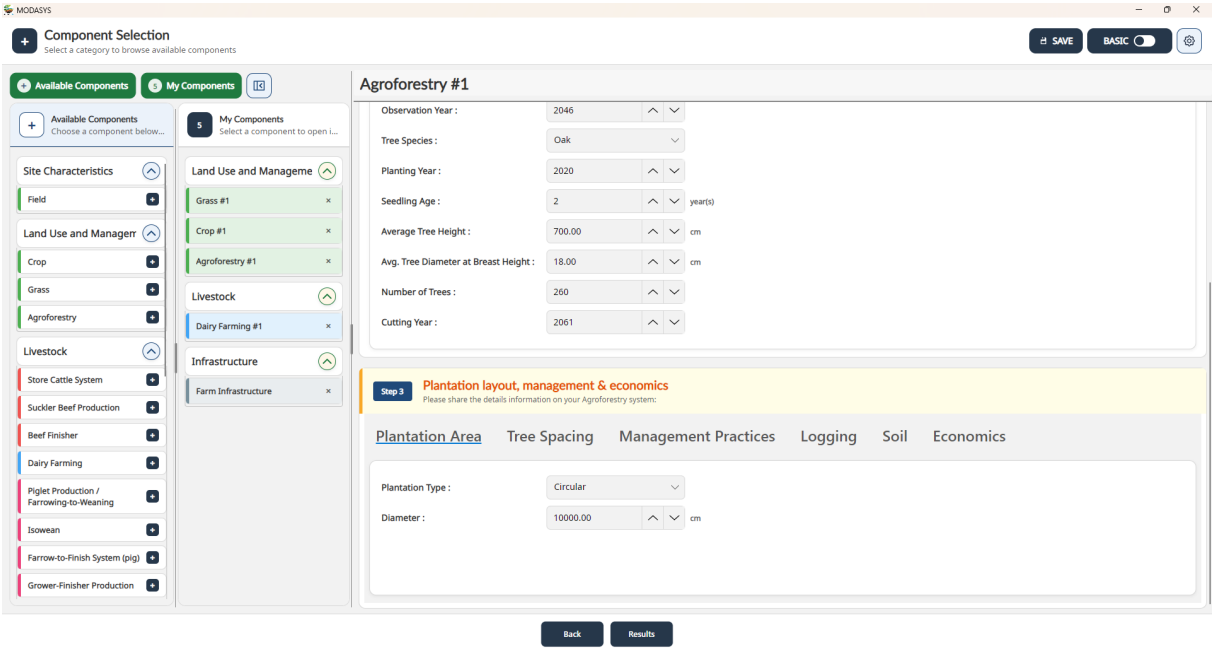



**Figure 5.25:** *Plantation Area tab.*

**Tree Spacing** The spacing between trees and rows.

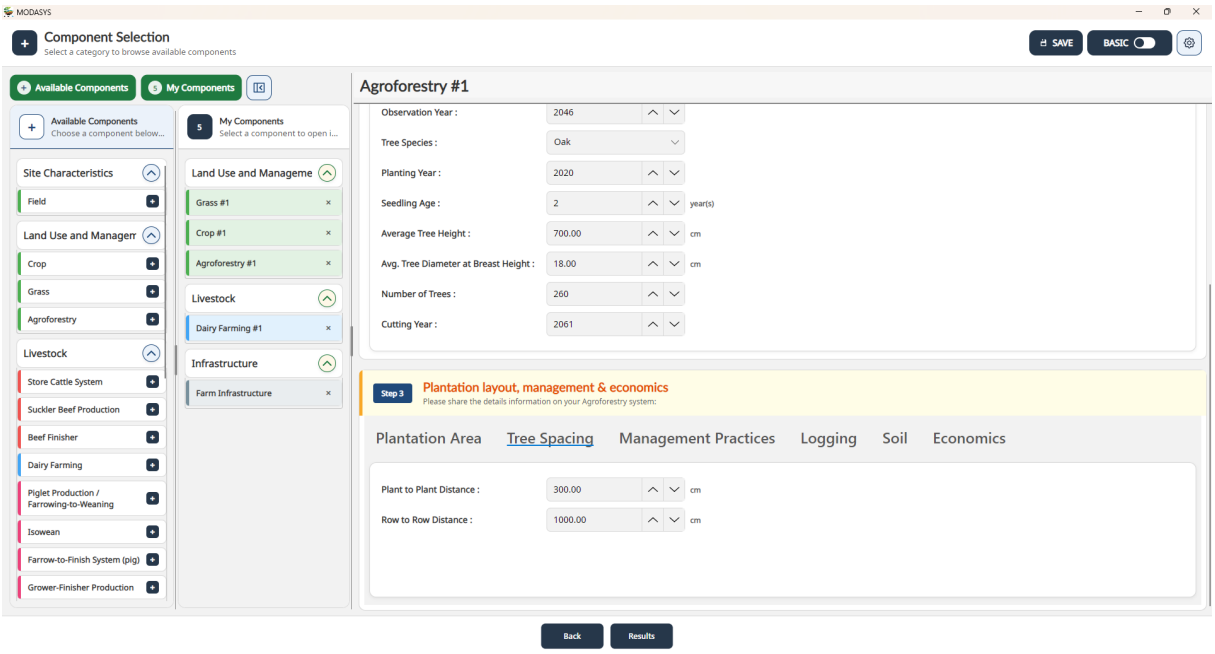



**Figure 5.26:** *Tree Spacing tab.*

**Management Practices** Plant management (thinning, crown pruning with the years it is done, coppicing, raising, vista pruning, reduction, pollarding, espalier, topiary), fertilization, irrigation and soil management (Figure 5.27).

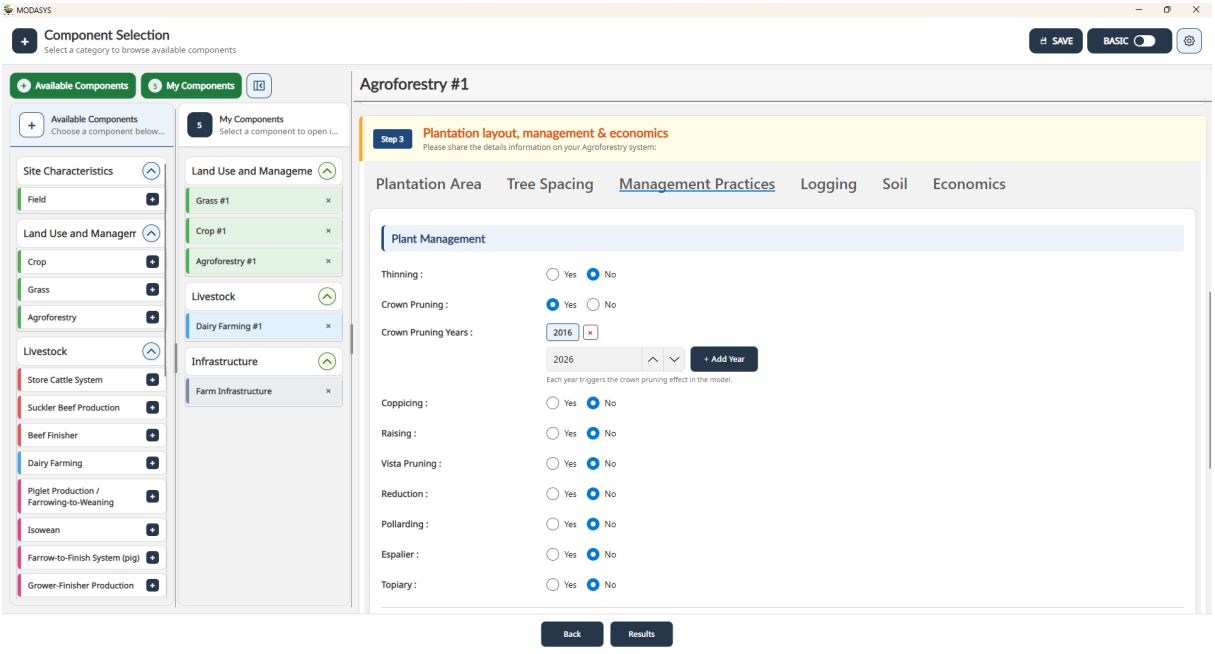



**Figure 5.27:** *Management Practices tab.*

**Logging** Whether logging takes place (Figure 5.28).

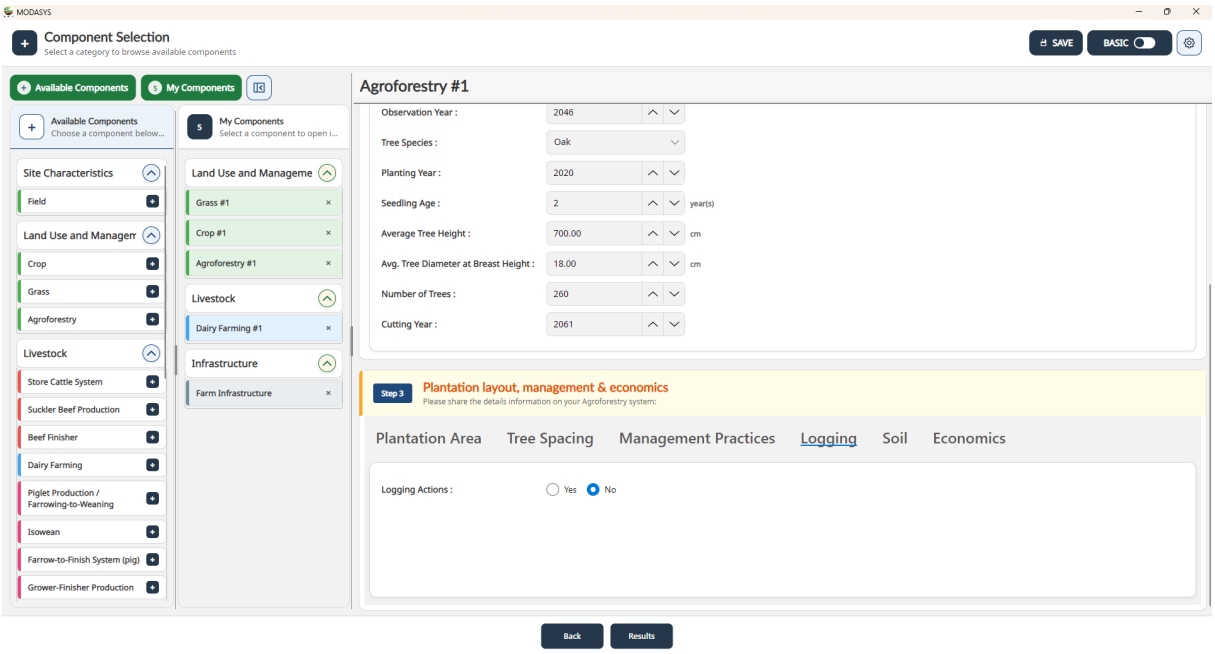



**Figure 5.28:** *Logging tab.*

**Soil and Economics** The soil under the trees, and the costs and prices of the agroforestry system.

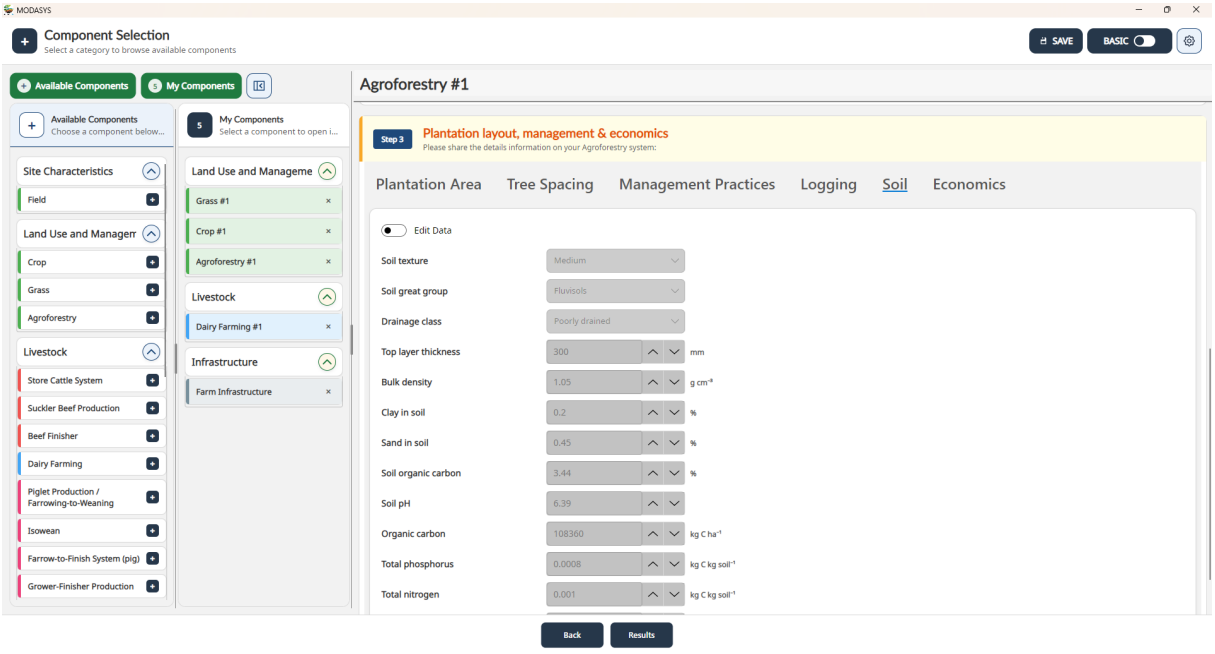



**Figure 5.29:** *Soil tab of the agroforestry component.*

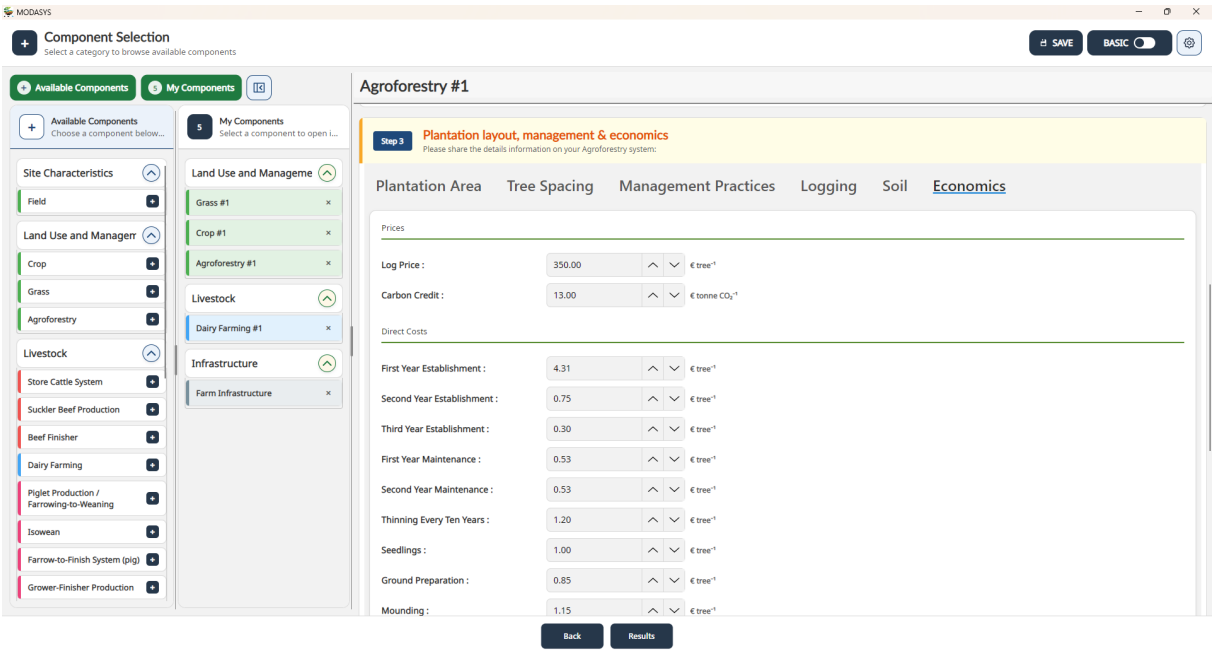



**Figure 5.30:** *Economics tab of the agroforestry component.*

## 5.6 Livestock Components (Dairy Example)

Livestock components follow the same pattern; the Dairy Farming component is shown here.

**Step 1 – Animal groups.** Add groups with + **Calves**, + **Heifers**, + **Lactating** and + **Bulls**, then select a group in the table. The table shows each group's type and number of management periods.

**Step 2 – Management periods.** The periods of the selected group, each with a name, start and end date, and length in days (for example housed lactation, grazing lactation and dry period, repeated each year). Click + **Add period** to add one, and select a period to edit it.

**Step 3 – Period details.** The tabs of the selected period. The box at the top right shows which period is being edited.

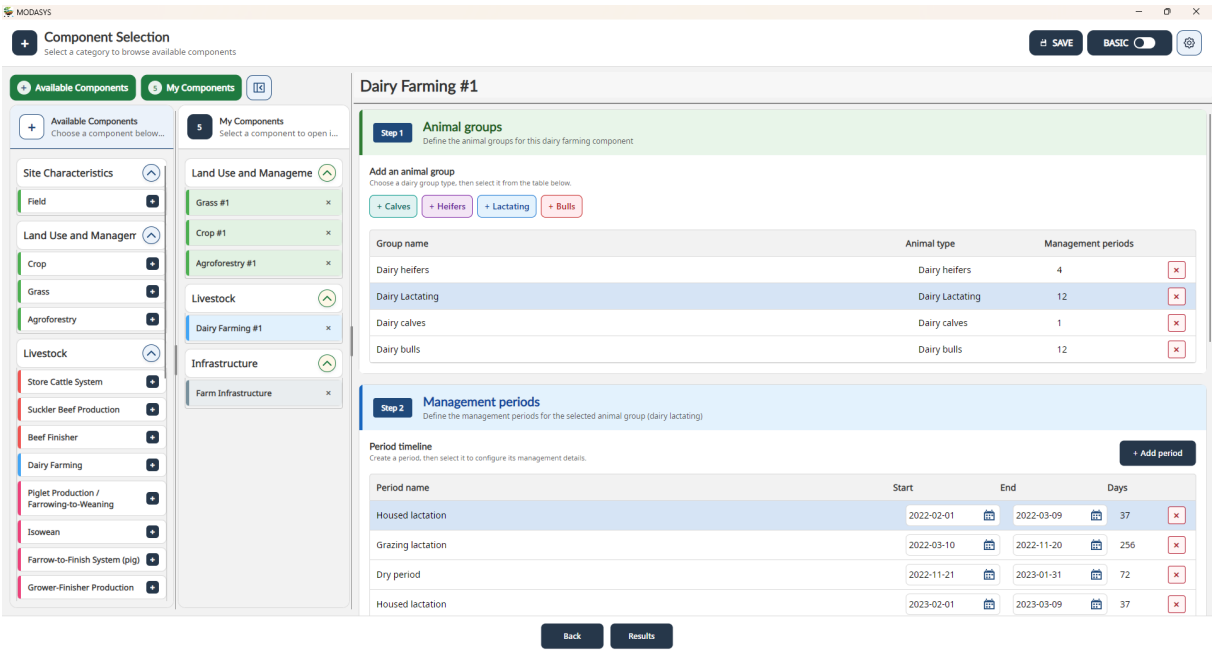



**Figure 5.31:** *Dairy Farming component: animal groups (Step 1) and the management periods of the lactating cows (Step 2).*

**General** Number of animals, milk production (kg day$^{-1}$), milk fat and protein content, and number of culled cows (Figure 5.32).

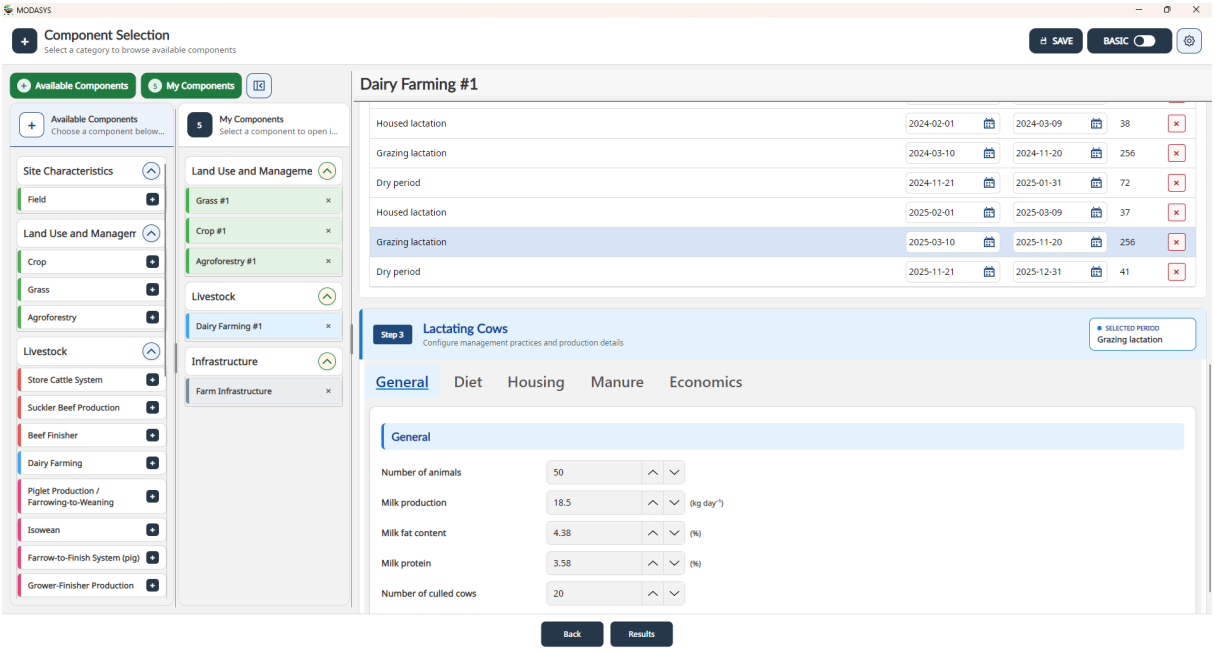



**Figure 5.32:** *General tab of a dairy management period.*

**Diet** The diet type and any diet additive. **Custom Diet Creator** opens a tool to formulate your own diet from feed ingredients (Figure 5.33).

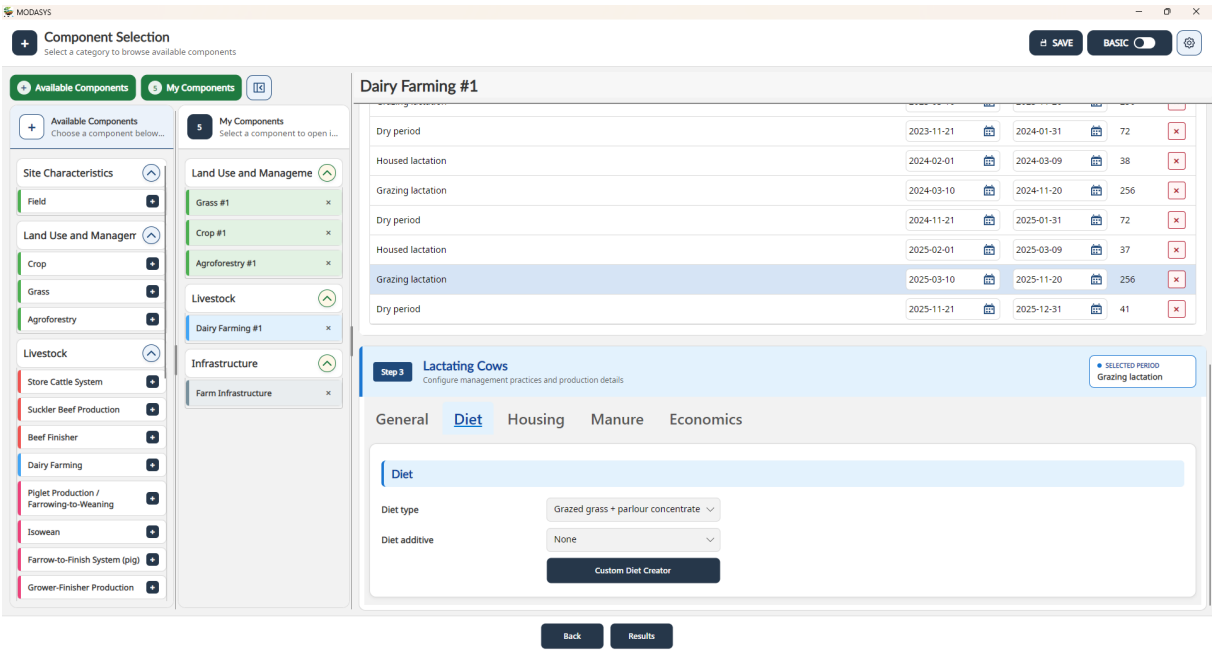



**Figure 5.33:** *Diet tab.*

**Housing** The housing type for the period. For *Pasture/range/paddock*, choose the grass field the animals graze; MODASYS reports the nitrogen in the dung and urine deposited on that pasture. This is what links the herd to the grass field's Grazing tab (Figure 5.34).

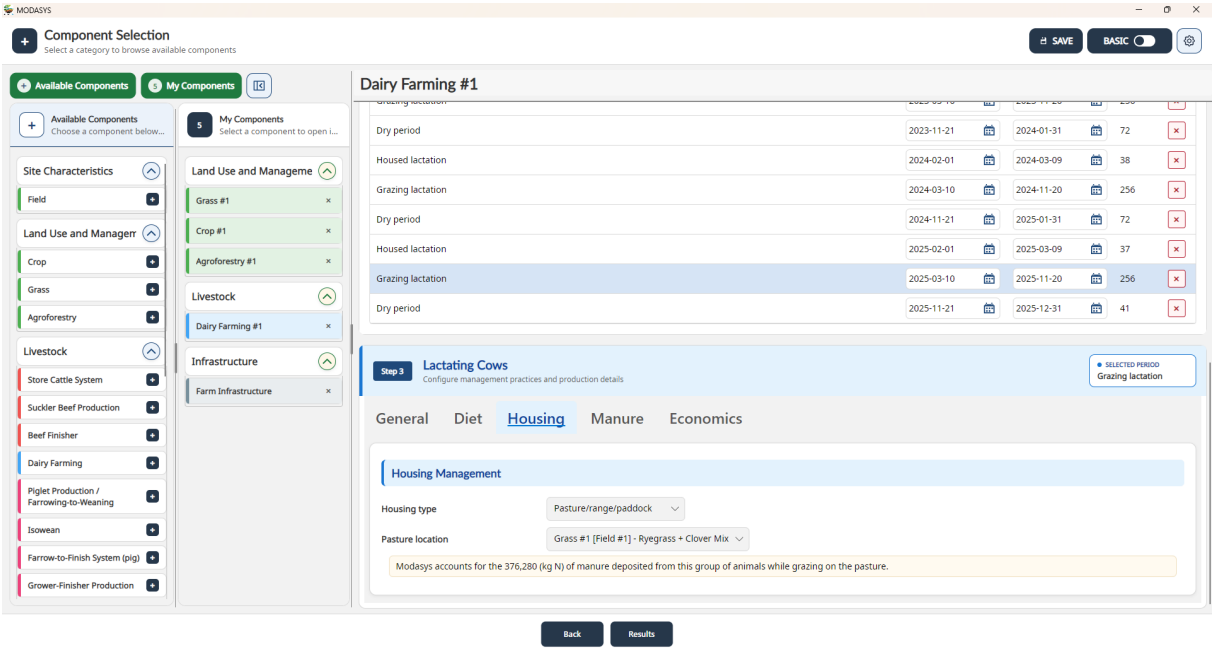



**Figure 5.34:** *Housing tab: the lactating cows graze Grass #1 during the grazing season.*

**Manure** The manure handling system of the period (e.g. liquid slurry with natural crust when housed, pasture when grazing) (Figure 5.35).

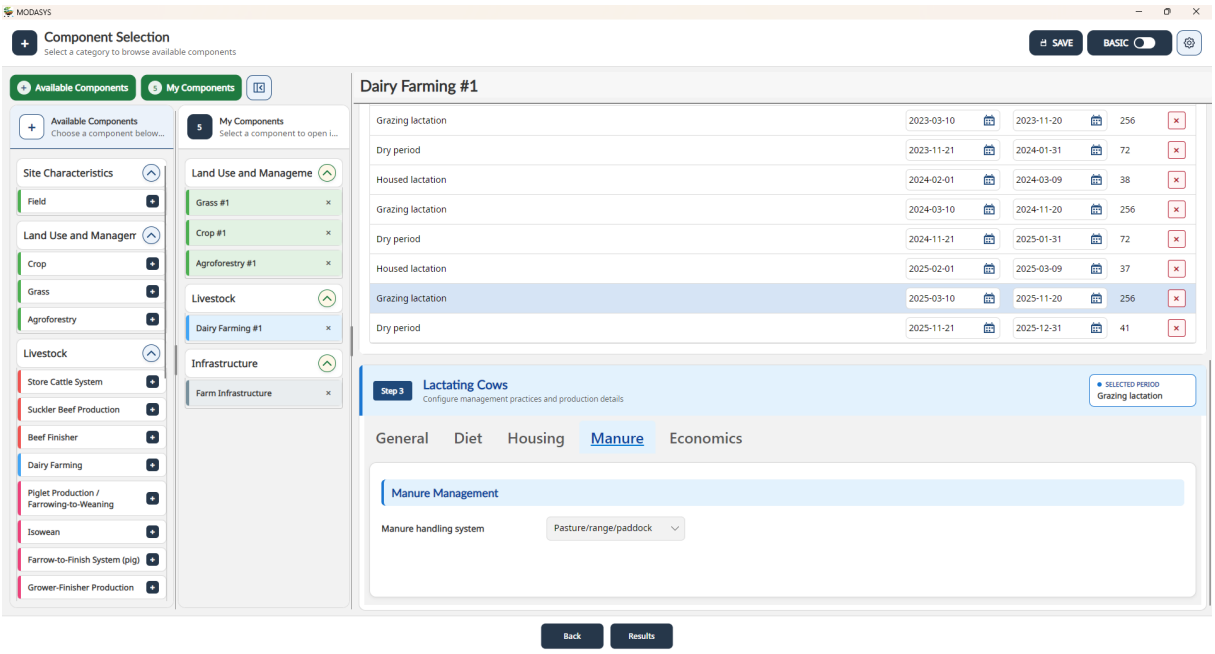



**Figure 5.35:** *Manure tab.*

**Economics** Costs and revenues of the group.

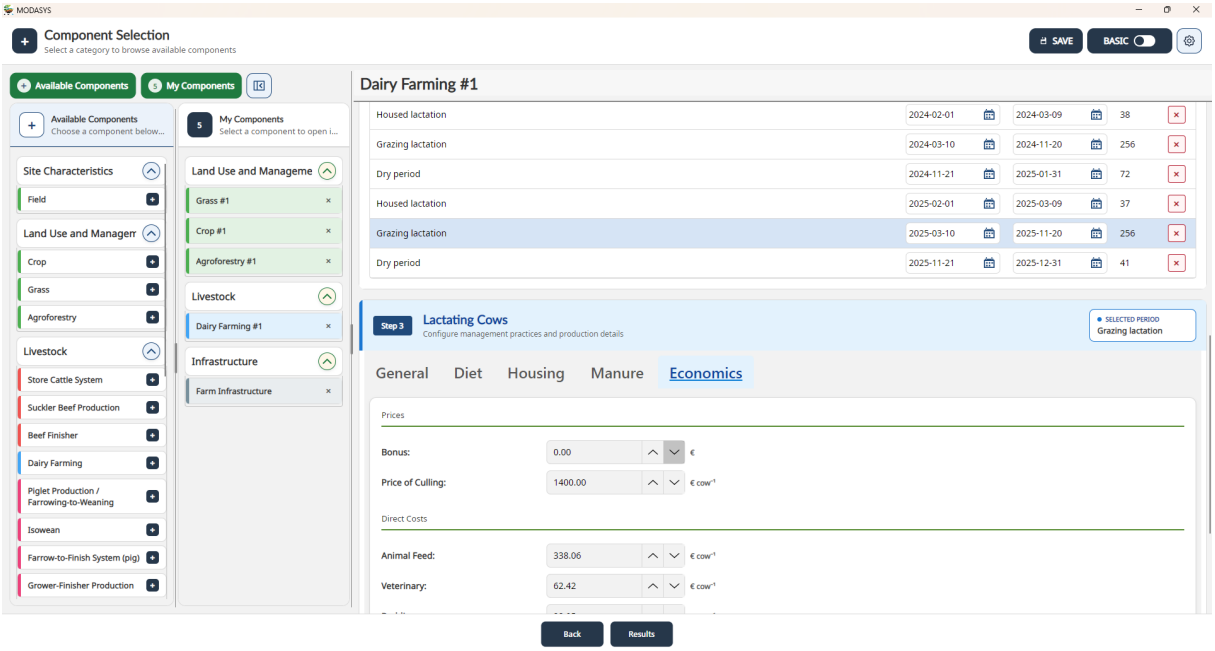



**Figure 5.36:** *Economics tab of a dairy management period.*

## 5.7 Farm Infrastructure Component

The Farm Infrastructure component holds the anaerobic digester and the farm's physical assets and annual energy flows. It has six steps:

1. **Digestate separation** – whether the digestate is separated into liquid and solid fractions.

2. **Manure sources** – tick the livestock management periods whose manure goes to the digester and the daily proportion added. Only housed periods produce manure that can be collected; manure deposited while grazing stays on the pasture.
3. **Farm residue substrates** – farm residues co-digested with the manure.
4. **Digestor output** – the target output (e.g. electricity production).
5. **Farm infrastructure assets** – buildings, stores, roads, machinery, energy and water systems and linear landscape features. For each asset give the type, quantity, lifetime, land occupation, linear length and whether the surface is sealed. Lifetime spreads the construction emissions over the asset's life; land occupation and linear length feed the biodiversity indicators.
6. **Annual operating flows** – diesel, electricity, heat and water used or produced in a year, with the origin (purchased or produced on the farm) and the life-cycle stage (upstream or downstream).

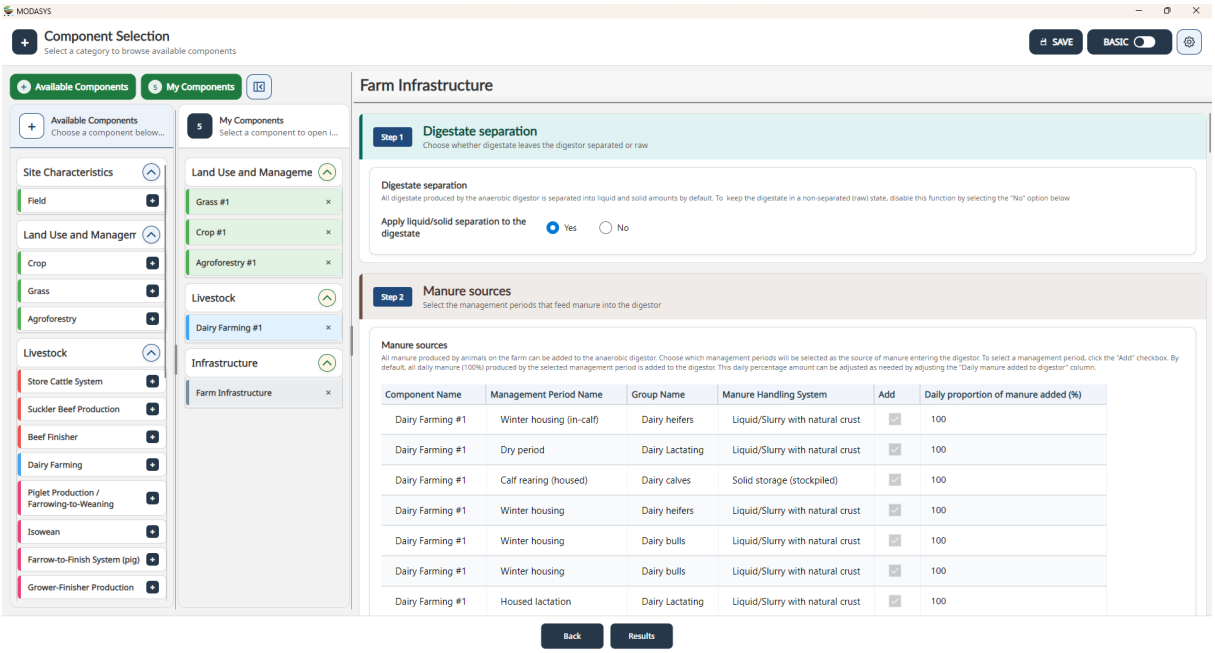



**Figure 5.37:** *Farm Infrastructure, Steps 1–2: digestate separation and the manure sources of the digester.*

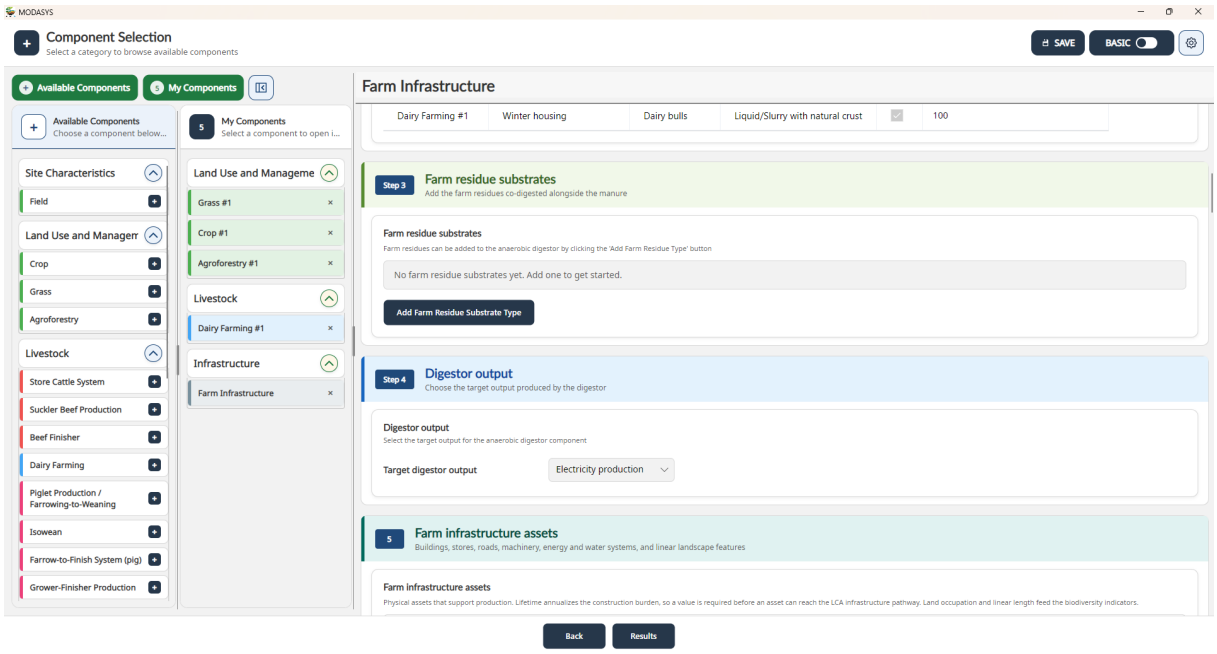



**Figure 5.38:** *Steps 3–4: farm residue substrates and digestor output.*

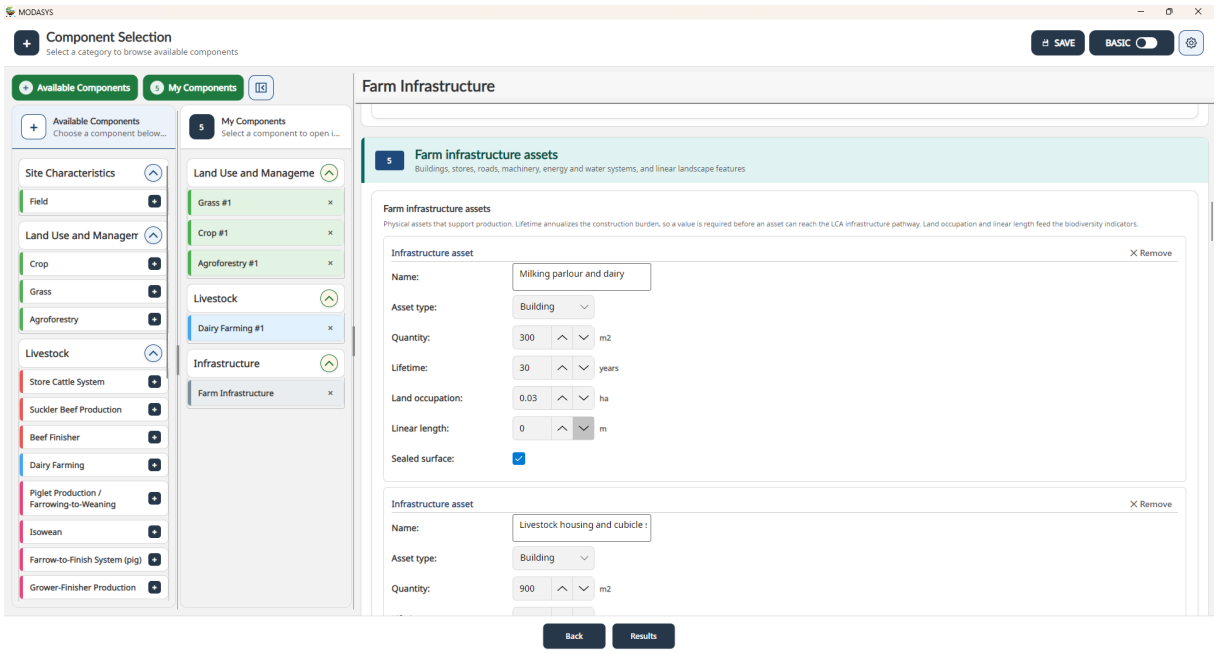



**Figure 5.39:** *Step 5: farm infrastructure assets.*

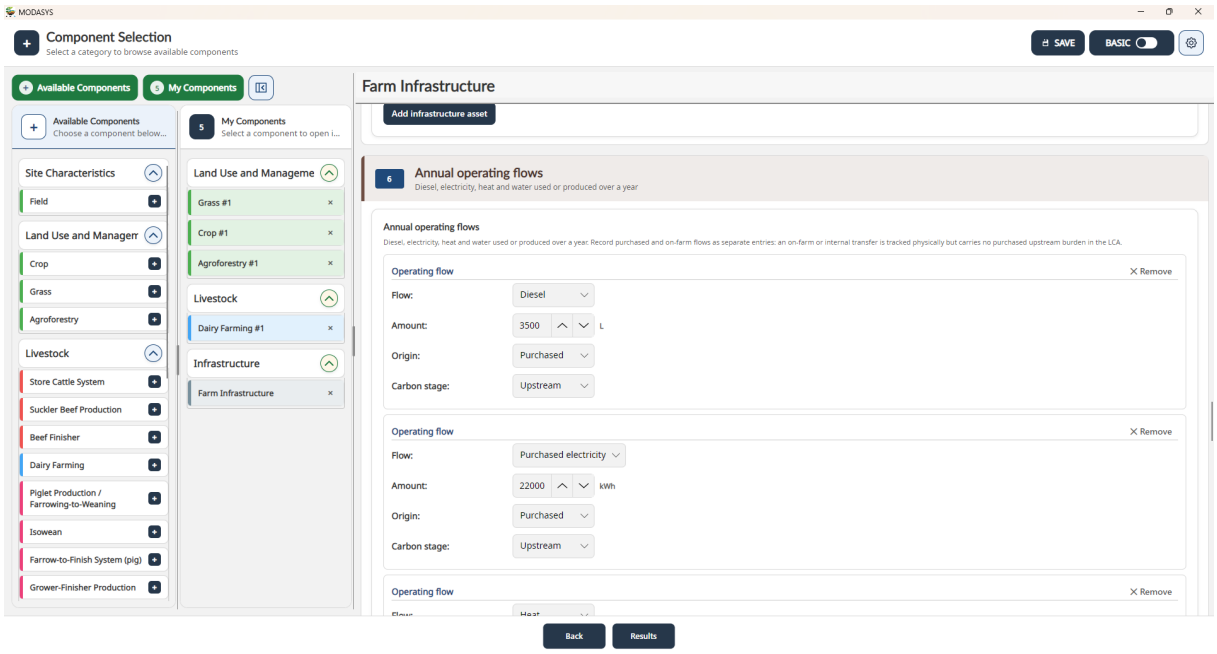



**Figure 5.40:** *Step 6: annual operating flows (diesel, purchased electricity, heat).*

# 6. Options and Settings

The gear icon at the top right of the Component Selection screen opens the **Options** page (Figure 6.1). Each entry opens its own page; **Back** returns to the list.

- **File:** New Farm, Open Farm, Close Farm, Manage Farm (the stored farms), Save, Save As, Import Farm, Export Farm and Export Climate.
- **Farm:** Farm Settings, User Settings, Temperature, Precipitation and Evapotranspiration.

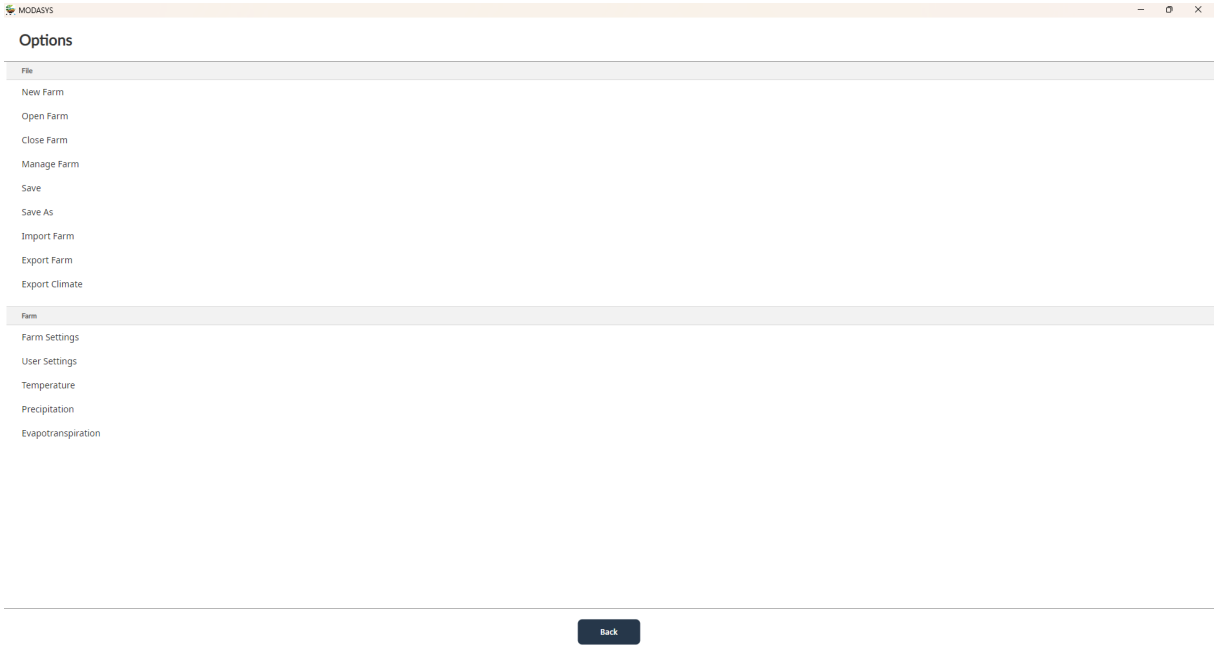



**Figure 6.1:** *The Options page.*

**Farm Settings** The farm's name and comments, its location (polygon, coordinates, country, hardiness zone), growing-season precipitation and evapotranspiration, ecoregion factor, farm diesel use and the **Operation Mode** toggle (Basic or Advanced, the same setting as the switch on the Component Selection screen) (Figure 6.2). To change the soil of a field, use the field's Soil tab.

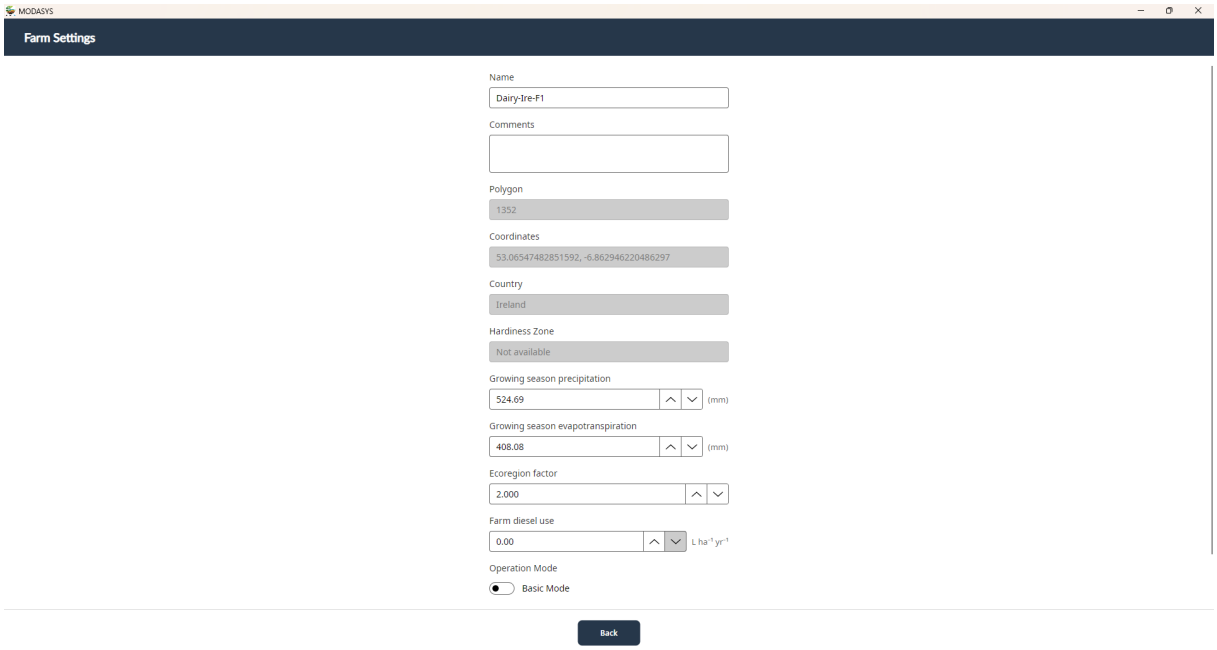



**Figure 6.2:** *Farm Settings of Dairy-Ire-F1 (polygon 1352).*

**User Settings** The carbon modelling strategy (e.g. IPCC Tier 2), run-in period and carbon concentration, and the switches that turn on the crop growth, grass growth and livestock growth models. With a growth model switched on, yields or animal performance are predicted from weather, soil, management and diet instead of being taken from your inputs (Figure 6.3).

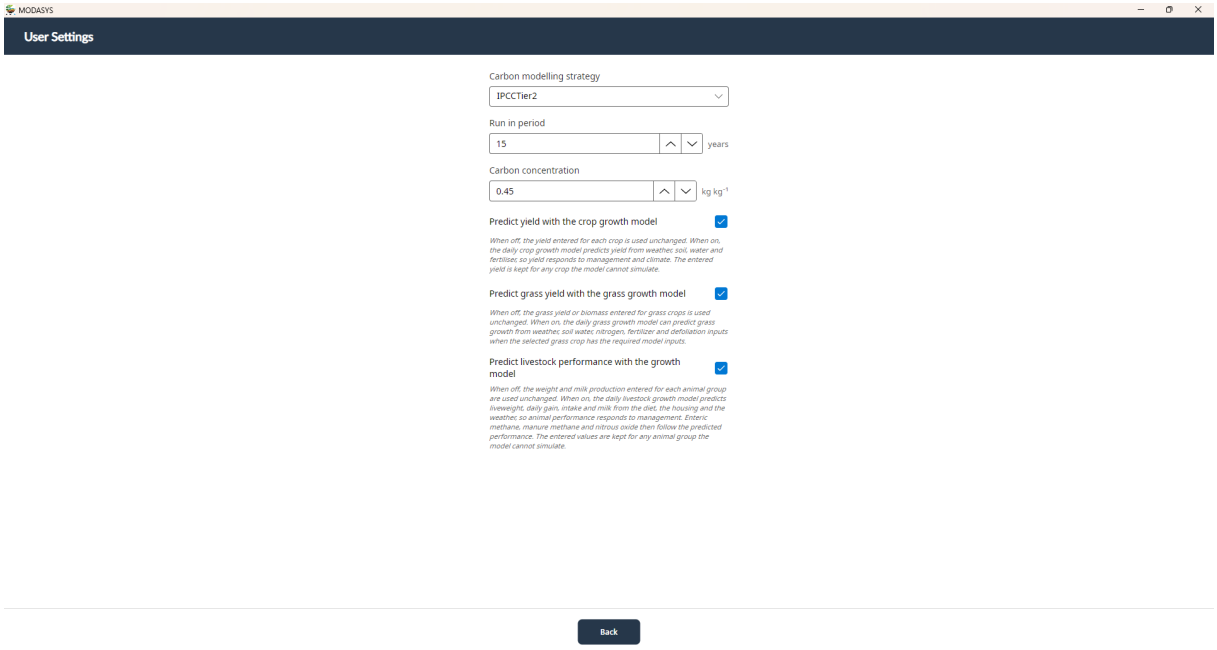



**Figure 6.3:** *User Settings, including the crop, grass and livestock growth model switches.*

**Temperature, Precipitation and Evapotranspiration** Monthly averages for the farm, each shown as a chart that you can edit month by month (Figures 6.4–6.6).

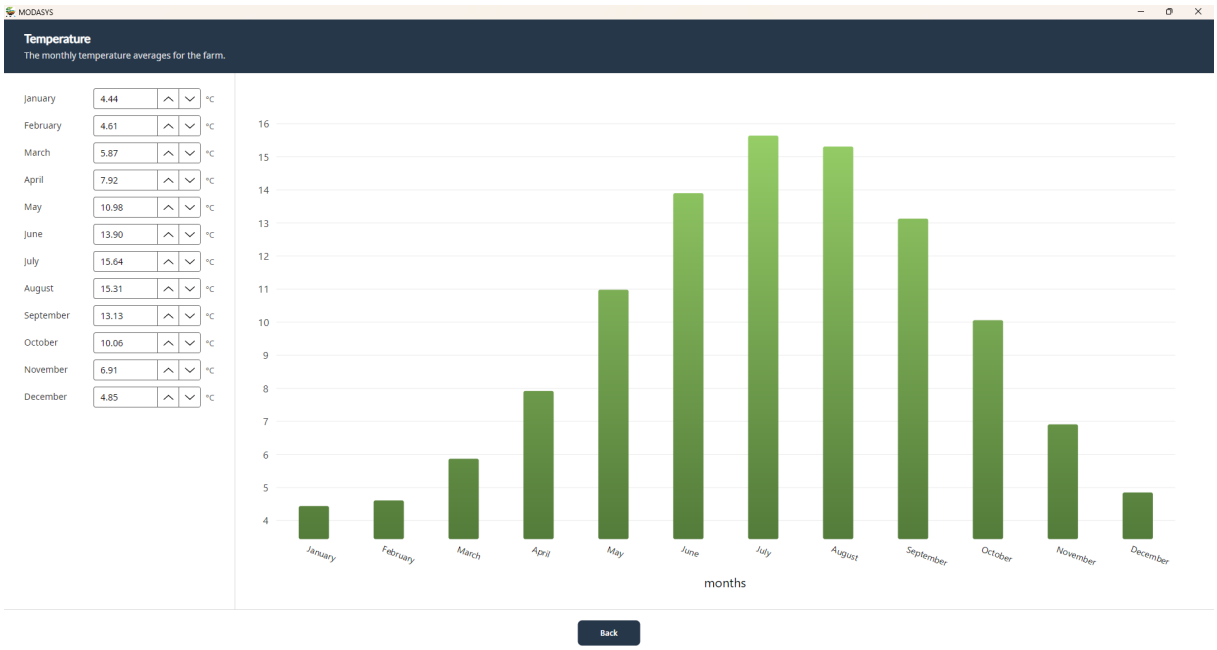



**Figure 6.4:** *Monthly temperature averages for the farm.*




**Figure 6.5:** *Monthly precipitation for the farm.*

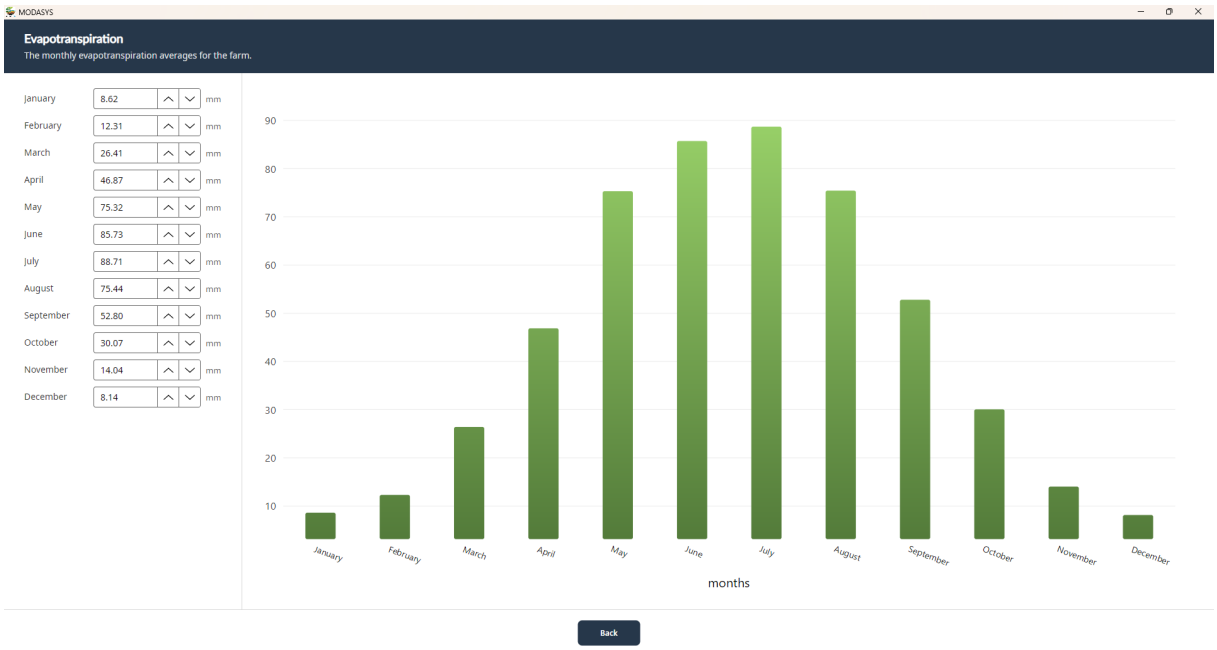



**Figure 6.6:** *Monthly evapotranspiration for the farm.*

# 7. Results

Click **Results** at the bottom of the Component Selection screen to run the model for the whole farm. The results screen (Figure 7.1) lists the result areas on the left; each area has one or more reports in tabs. **Recalculate** (top right) runs the model again after inputs have changed, and **Export to Excel** saves all results to a workbook. Most reports also have their own **Export to Excel** button and a **Table/Chart** switch. **Back** returns to the components.

**Table 7.1:** *Result areas and their reports.*

| Area | Reports |
|---|---|
| Production | All Productions, Feed Intake |
| Organic Carbon | Soil Organic Carbon, Biomass Carbon |
| Emissions | Detailed Emissions, Component Emissions, Sectoral Emissions, Monthly Variations |
| Energy | Field Operations, Infrastructure, Upstream, Downstream, AD Performance, Summary |
| Carbon Footprint | Total Carbon Balance, Net Carbon Balance |
| Indices | Soil Health Index, Biodiversity Index |
| Economics | Economics |

**Reading the charts.** Several reports share the same controls:

- **Years** – the range of years included. **View** – *All years (mean ± error)* shows the mean of the years that have a value; *Single year* shows one year.
- **Basis** – *Annual total* (the farm's actual emissions), *Per hectare* (land intensity; livestock has no per-hectare form) or *Per head* (each herd's emissions divided by its average number of animals).

- **Units** – $CO_2e$ (the default; one chart, using 100-year global warming potentials of 28 for $CH_4$ and 265 for $N_2O$) or *Gas mass* (kg of each gas, one chart per gas).
- **Error bar** – standard error (SE) or standard deviation (SD) across years or fields. Error bars need at least three values. The spread describes variation between years or fields, not model uncertainty.
- A **values-by-year table** below each chart lists the numbers behind it.

**Sign convention.** Positive values are emissions (a source). Negative values are carbon storage (a sink), for example soil carbon gain or tree growth.

## 7.1 Production

**All Productions** reports the annual (or monthly) crop yield and above- and below-ground biomass, grass production and livestock output (Figures 7.1 and 7.2). **Feed Intake** reports the average and total feed intake of each animal group (Figure 7.3). Where the crop, grass or livestock growth models are switched on, these values are predicted by the models.

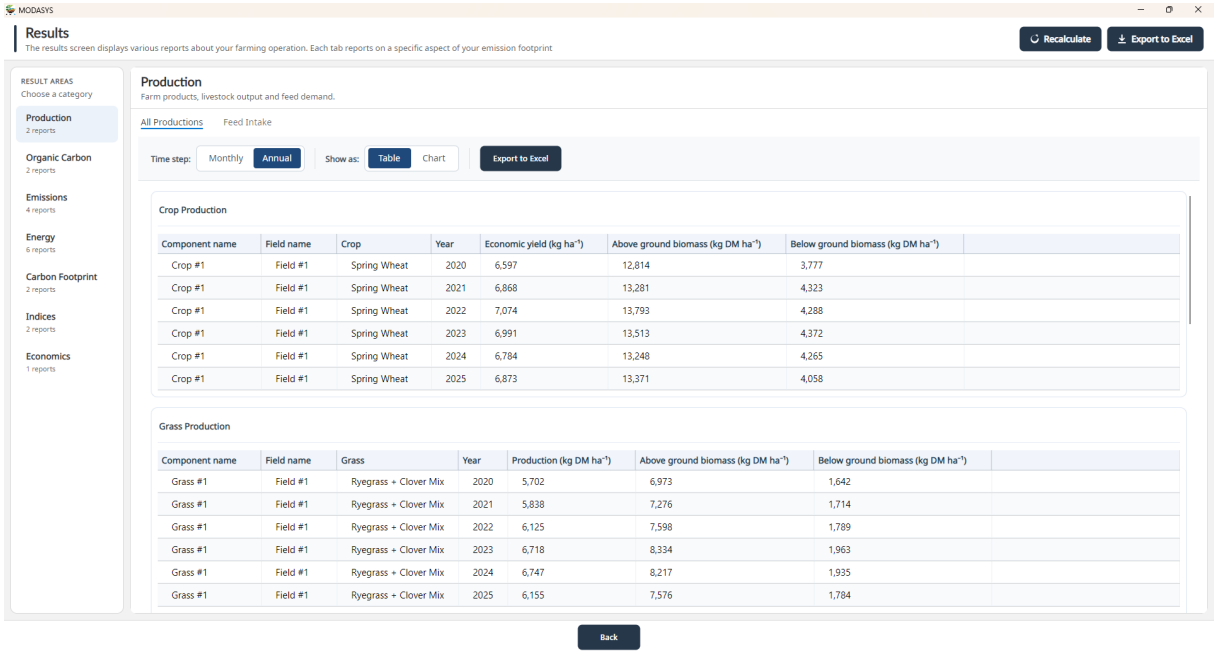



**Figure 7.1:** *The results screen. Result areas are on the left; the Production area shows crop and grass production by year.*

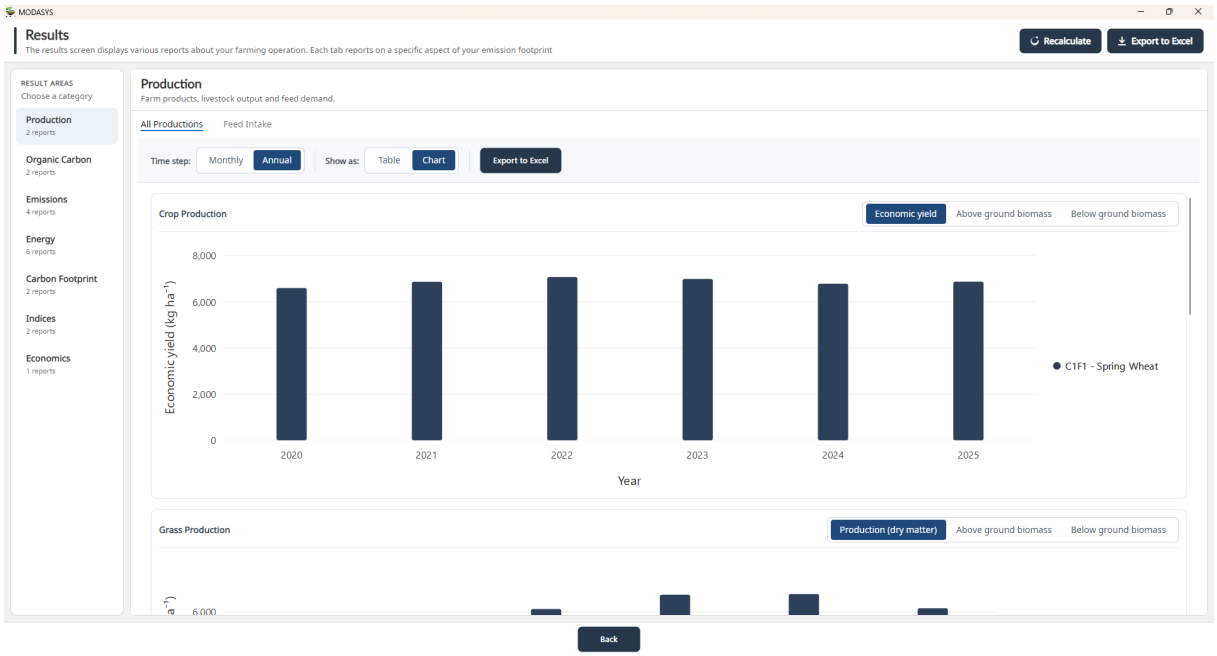



**Figure 7.2:** *All Productions shown as a chart. The buttons above each chart switch between yield and above- or below-ground biomass.*

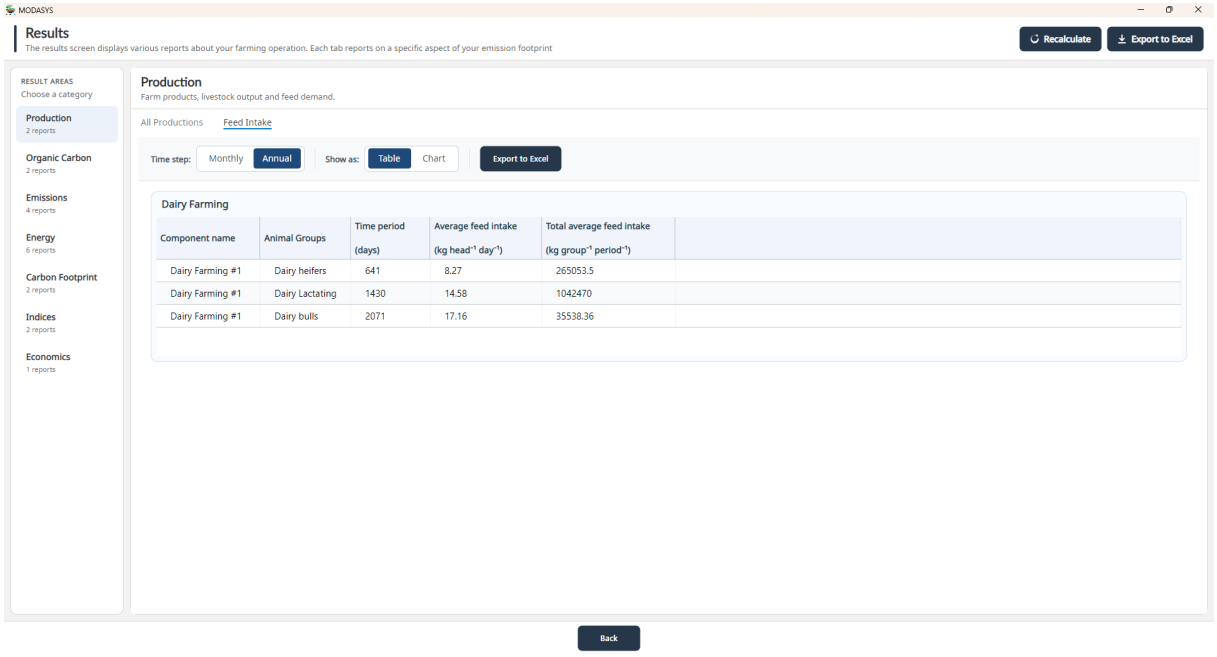



**Figure 7.3:** *Feed Intake of each dairy group.*

## 7.2 Organic Carbon

**Soil Organic Carbon** shows the soil organic carbon stock (kg C $ha^{-1}$) of each field over time, with one line per field, for crops and grass and for agroforestry (Figure 7.4). Comparing the lines shows how different management affects soil carbon. **Biomass Carbon** reports the above- and below-ground biomass carbon of crops, grass and agroforestry trees (Figure 7.5).

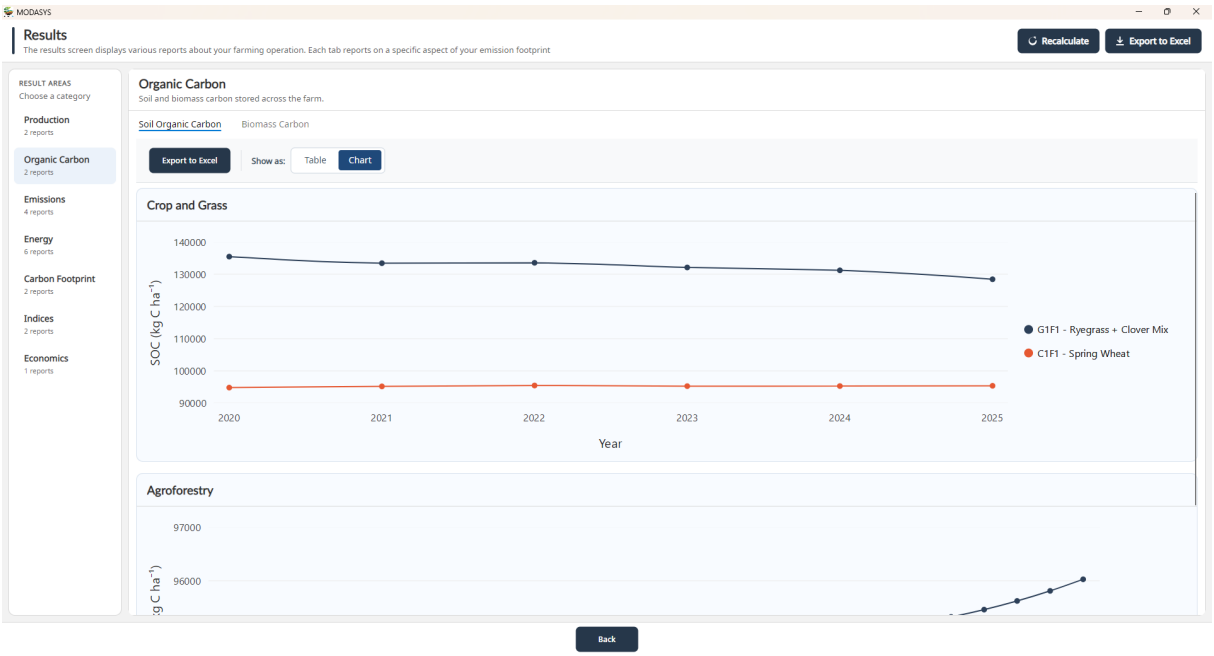



**Figure 7.4:** *Soil organic carbon of the grass field (G1F1) and crop field (C1F1), with agroforestry below.*

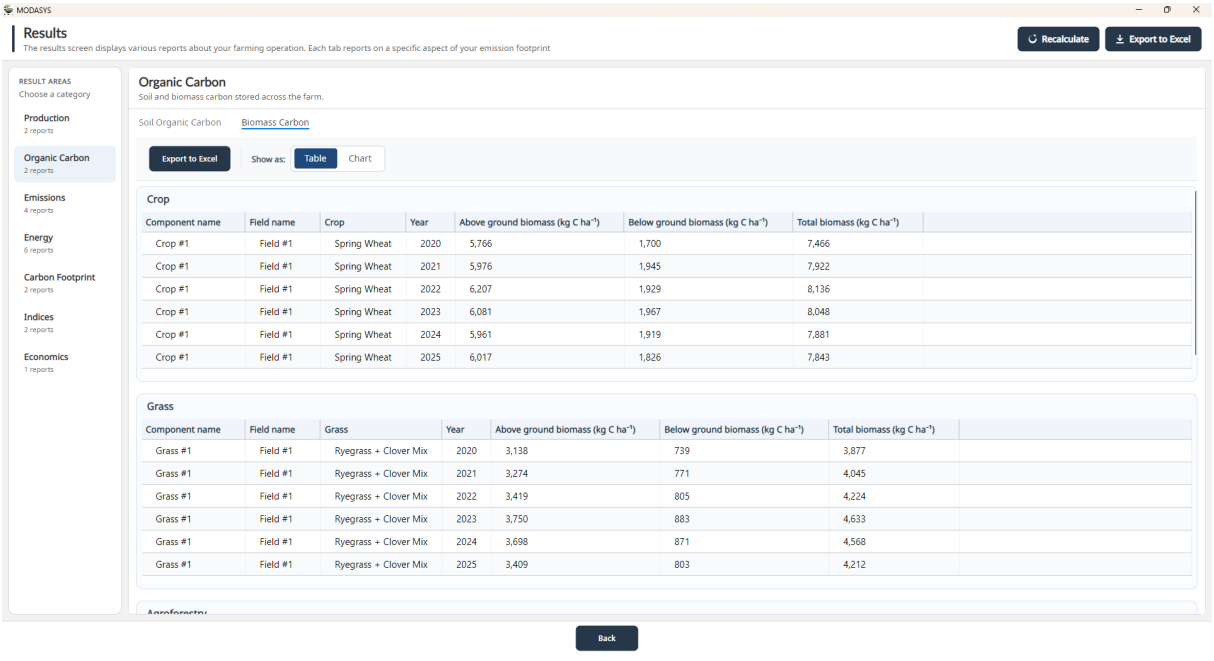



**Figure 7.5:** *Biomass carbon by field and year.*

## 7.3 Emissions

**Detailed Emissions** A table of every emission source of every component (Figure 7.6): enteric $CH_4$, manure $CH_4$, direct and indirect $N_2O$, farm energy $CO_2$ and upstream $CO_2$, with subtotals per component and a farm total. Choose the time step (monthly or annual), the basis (per ha or total), the report year and the units. The uncertainty of each emission type is shown below the table.

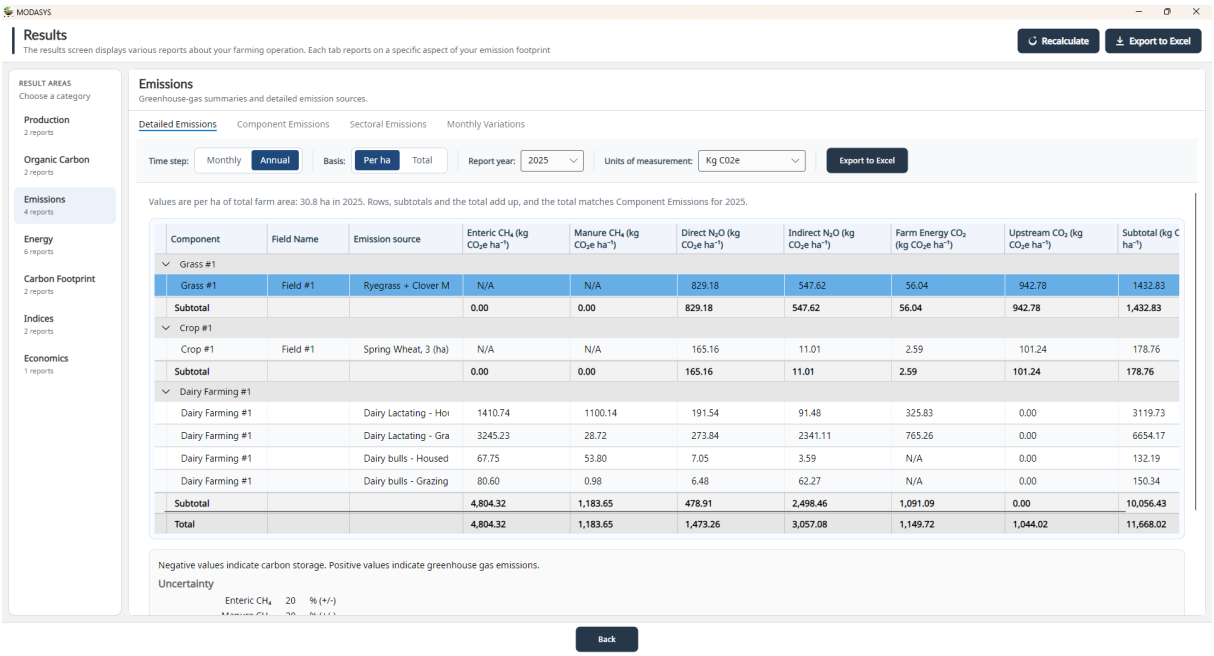



**Figure 7.6:** *Detailed Emissions for 2025, per hectare of farm area, in kg $CO_2e$.*

**Component Emissions** The farm's emissions by source (enteric $CH_4$, manure $CH_4$, manure-management $N_2O$, managed-soil $N_2O$, farm energy and so on), in $CO_2e$ on one chart; switch **Units** to *Gas mass* for one chart per gas (Figure 7.7).

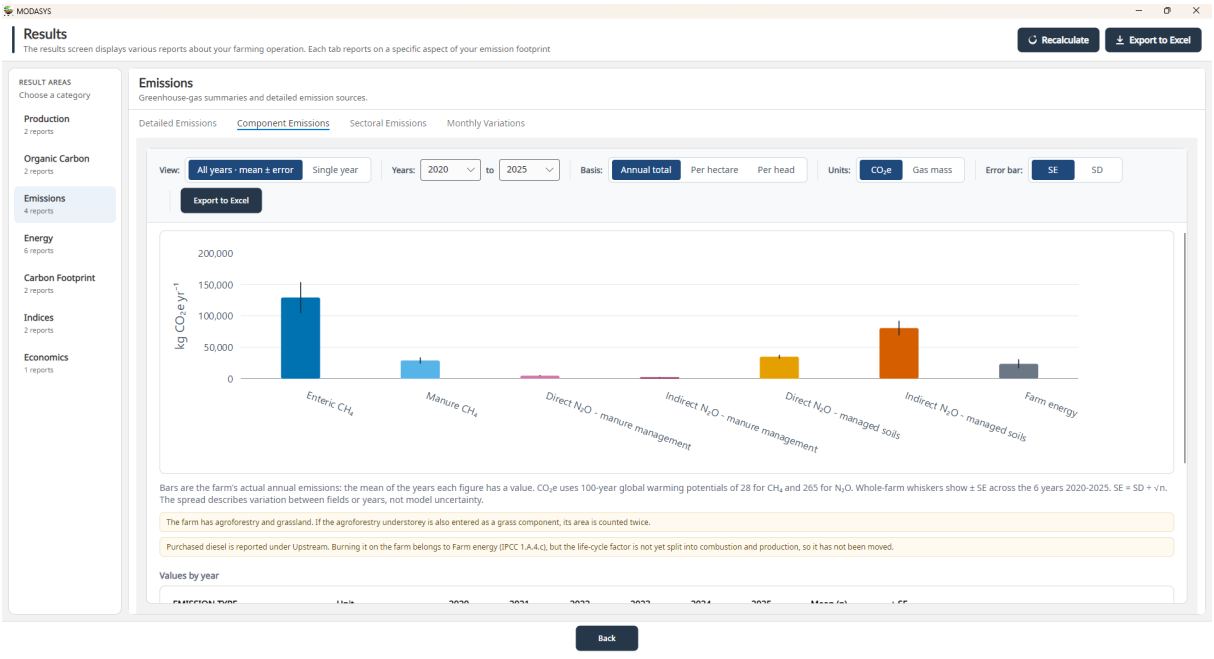



**Figure 7.7:** *Component Emissions: mean annual emissions by source in $CO_2e$, 2020–2025, with standard-error bars.*

**Sectoral Emissions** The farm's emissions by sector: livestock, cropland, grassland, agro-forestry and farm energy. In $CO_2e$ (the default) the chart is split into three life-cycle stages (Figure 7.8): **A – Before the farm gate** (one Upstream bar: production of fertiliser and herbicide, purchased farm inputs and the embodied emissions of infrastructure), **B – On the**

**farm** (the farm's own emissions, by sector) and **C – After the farm gate** (one Downstream bar: transport and processing). The farm footprint is cradle-to-farm-gate (A + B); downstream is shown for information only. Notes under the chart flag anything that needs attention, such as possible double counting of land. In gas mass the life-cycle stages are not shown, and each gas has its own chart (Figure 7.9).

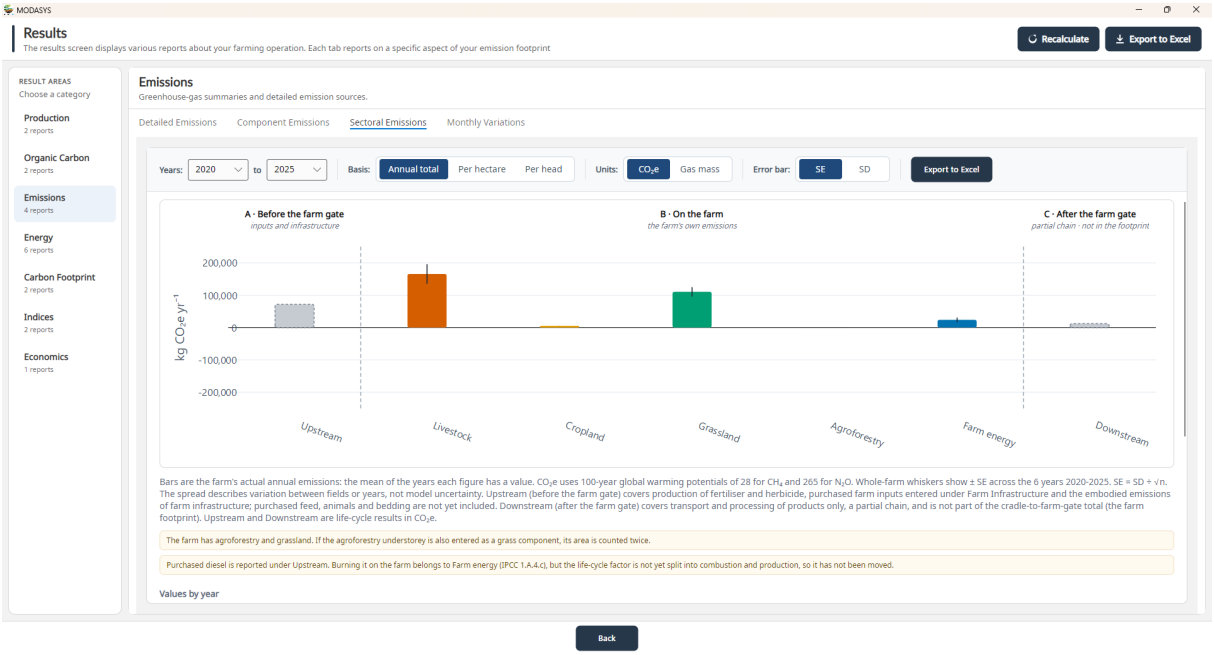



**Figure 7.8:** *Sectoral Emissions in $CO_2e$, with the three life-cycle stages.*

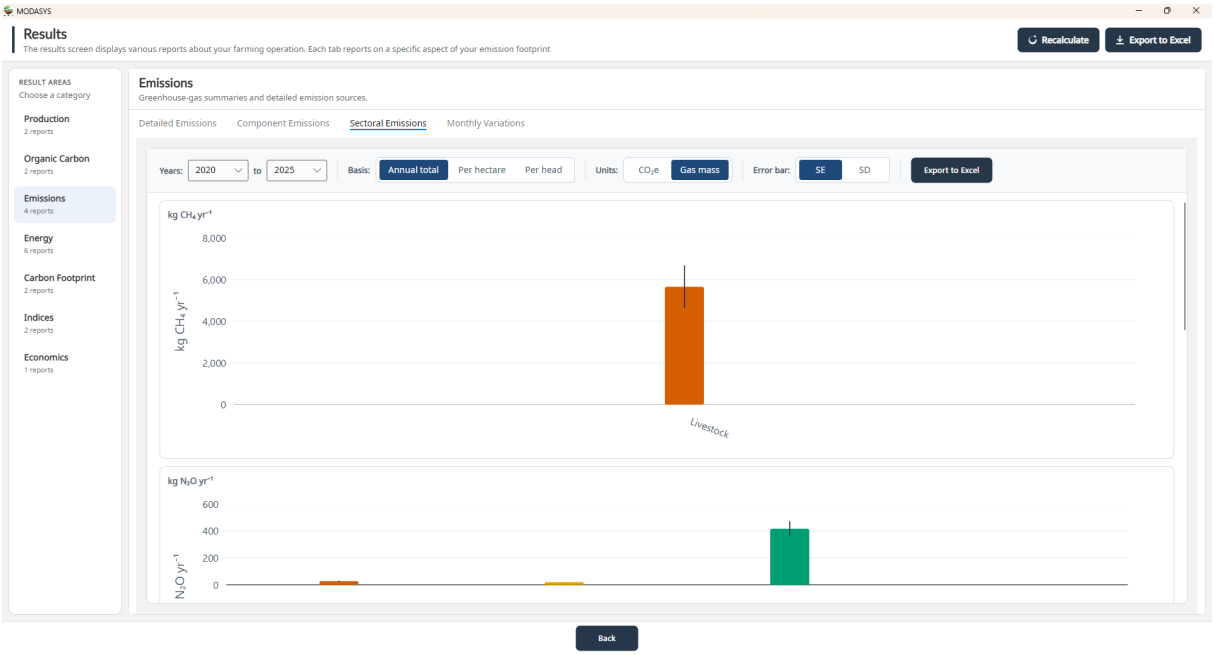



**Figure 7.9:** *Sectoral Emissions in gas mass: $CH_4$ (top) and $N_2O$ (bottom) by sector.*

**Monthly Variations** Emissions month by month, by emission type, showing the seasonal pattern, for example higher manure $CH_4$ during the housing months (Figure 7.10). Use **Show** to select emission types.

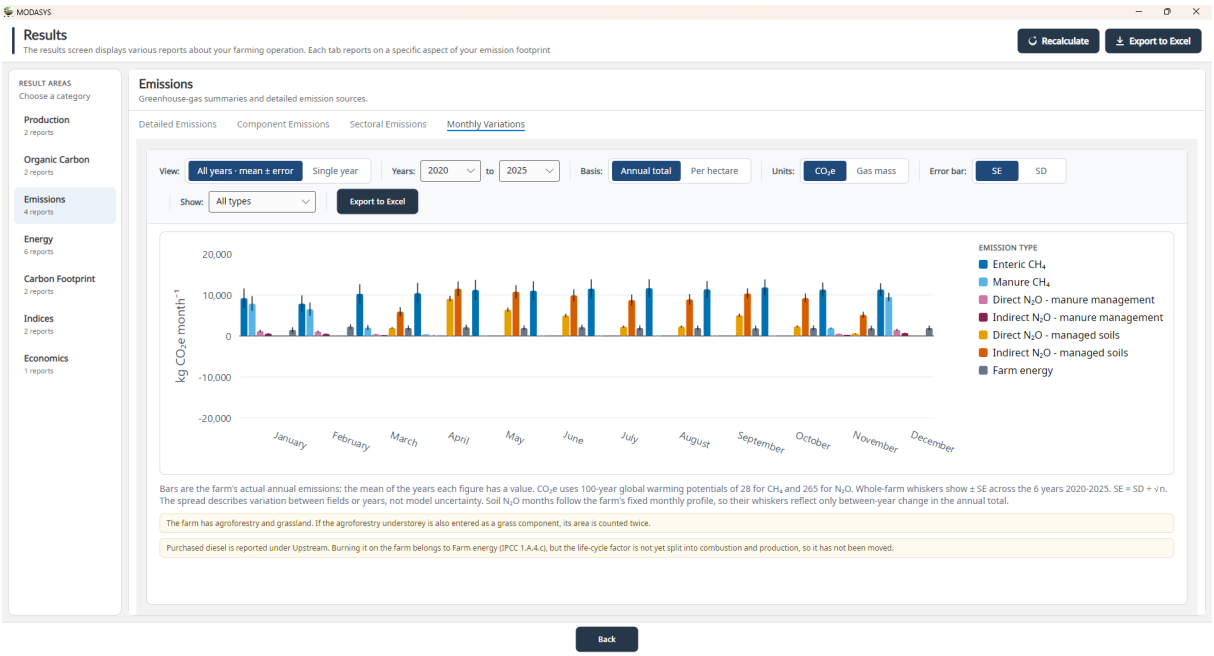



**Figure 7.10:** *Monthly Variations in $CO_2e$, by emission type.*

## 7.4 Energy

The Energy area reports energy-related and life-cycle emissions in kg $CO_2e$ $yr^{-1}$. Each report shows a headline figure, its share of the total on the Summary tab, the data status and the underlying records.

- **Field Operations** – energy used working the fields: fuel for tillage, sowing, spraying and harvest, fertilizer and manure application, irrigation and liming (Figure 7.11).
- **Infrastructure** – buildings, machinery and other assets, spread over their service life (Figure 7.12).
- **Upstream** – purchased energy, fuel, water, materials and services before they reach the farm (Figure 7.13).
- **Downstream** – transport, processing, export or use after products leave the farm (Figure 7.14).
- **AD Performance** – annual (or, with **Time step**, monthly) biogas, methane, electricity and heat produced by the anaerobic digester (Figure 7.15).
- **Summary** – the farm's energy footprint at a glance: assets and flows recorded, total carbon assessed, the largest source, and warnings about any data that needs attention (Figure 7.16).

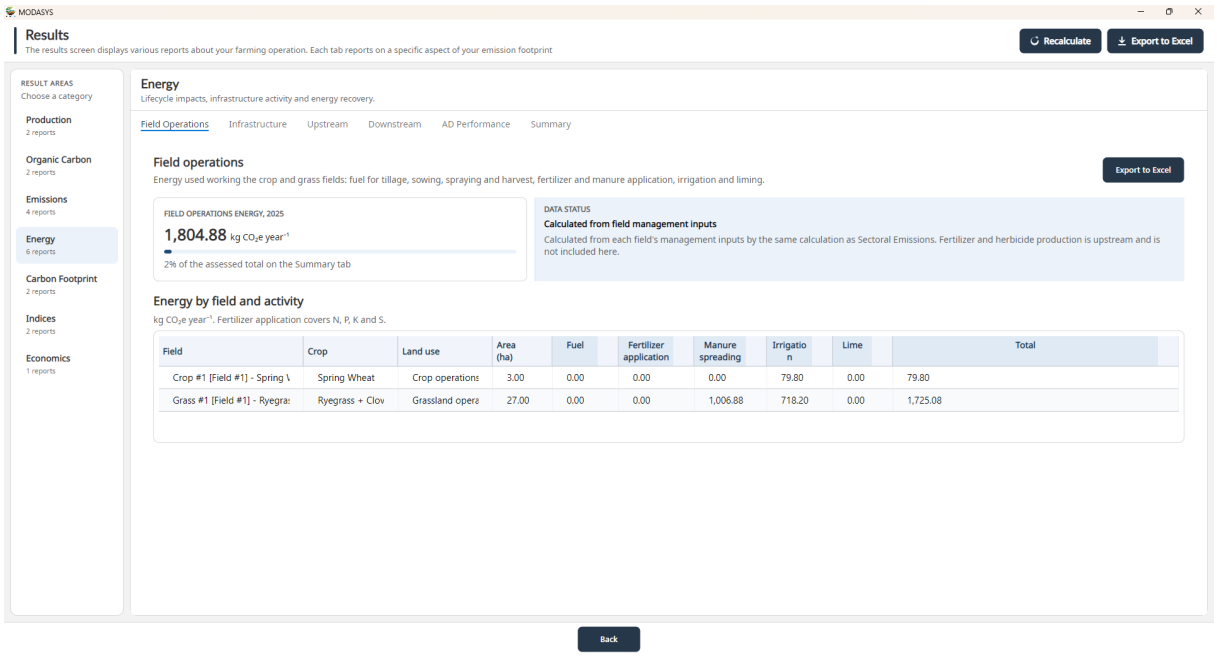



**Figure 7.11:** *Energy – Field Operations.*

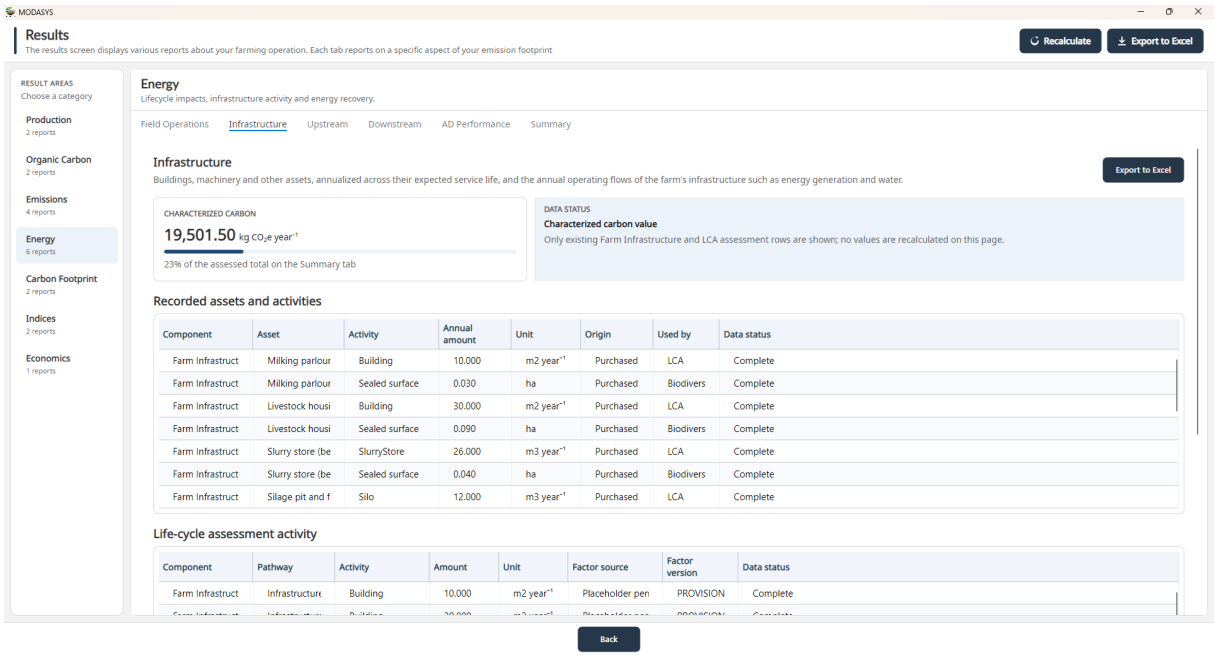



**Figure 7.12:** *Energy – Infrastructure.*

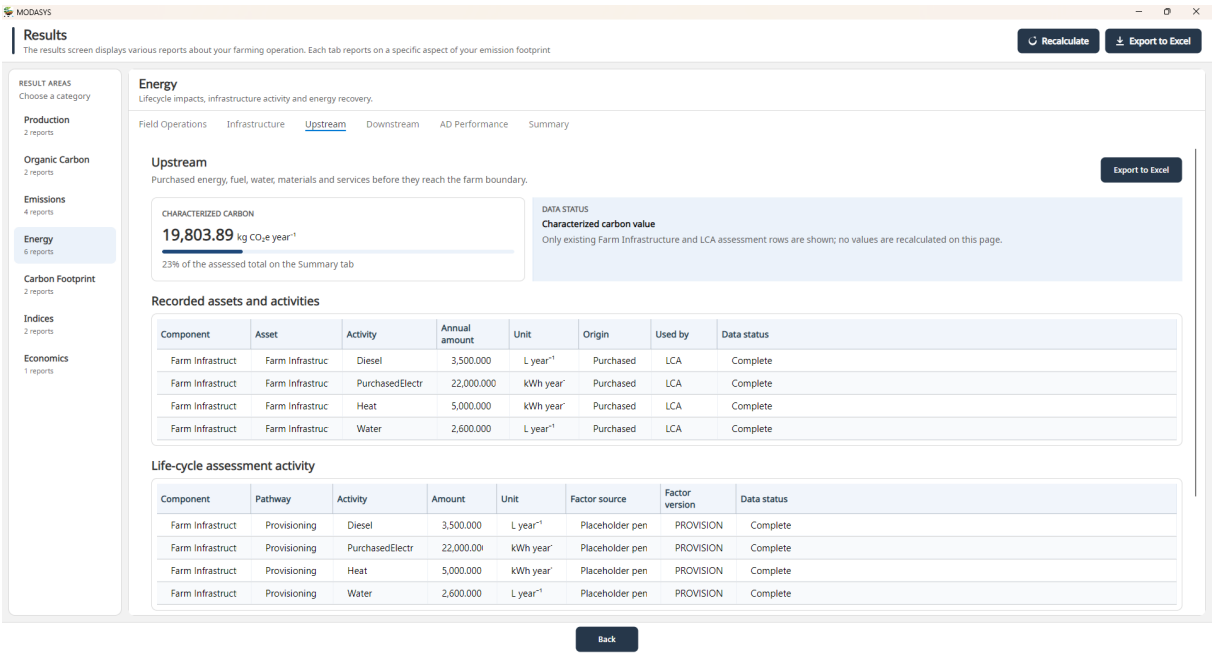



**Figure 7.13:** *Energy – Upstream.*

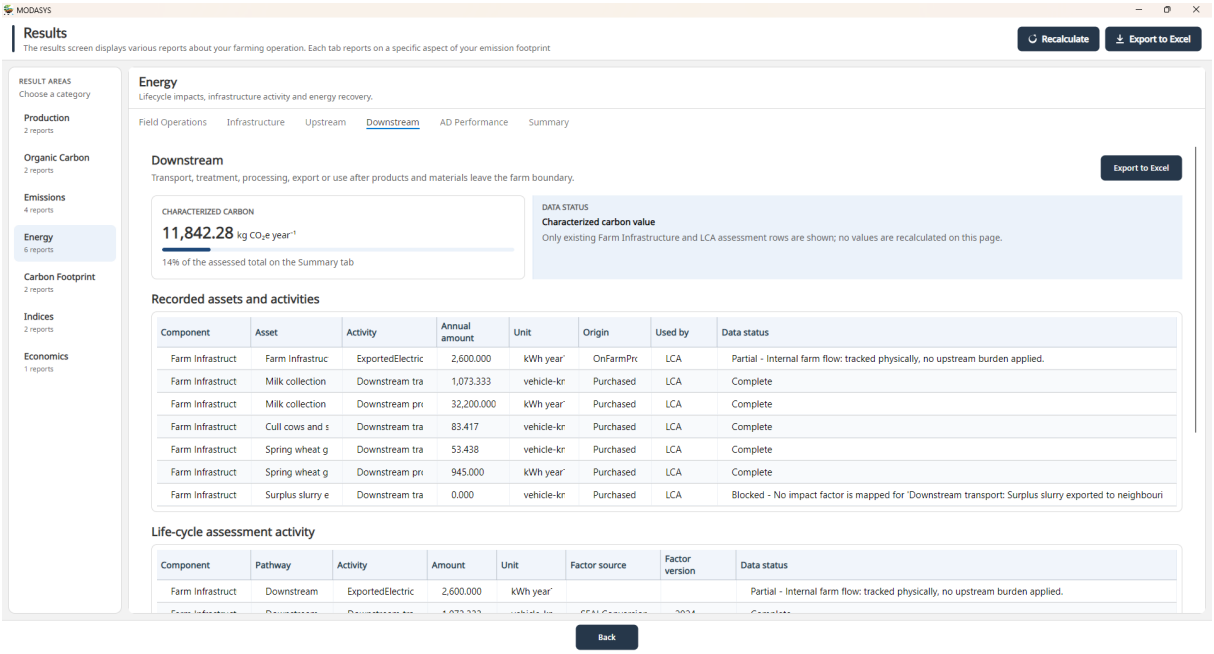



**Figure 7.14:** *Energy – Downstream.*

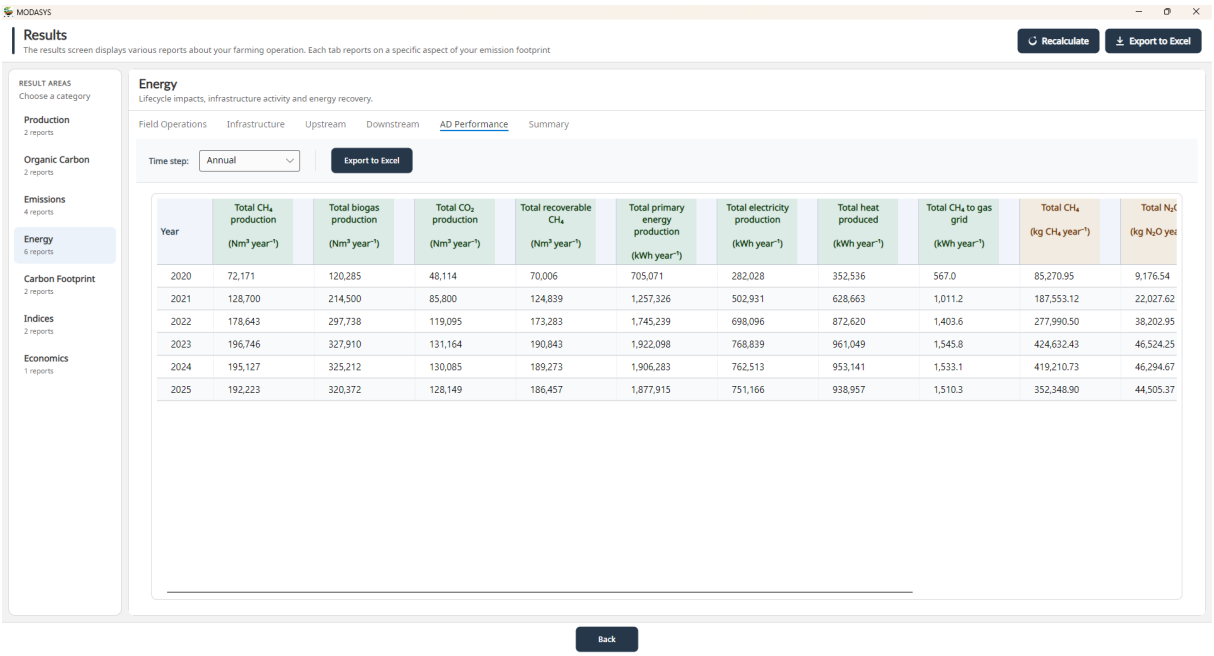



**Figure 7.15:** *Energy – AD Performance of the anaerobic digester, by year.*

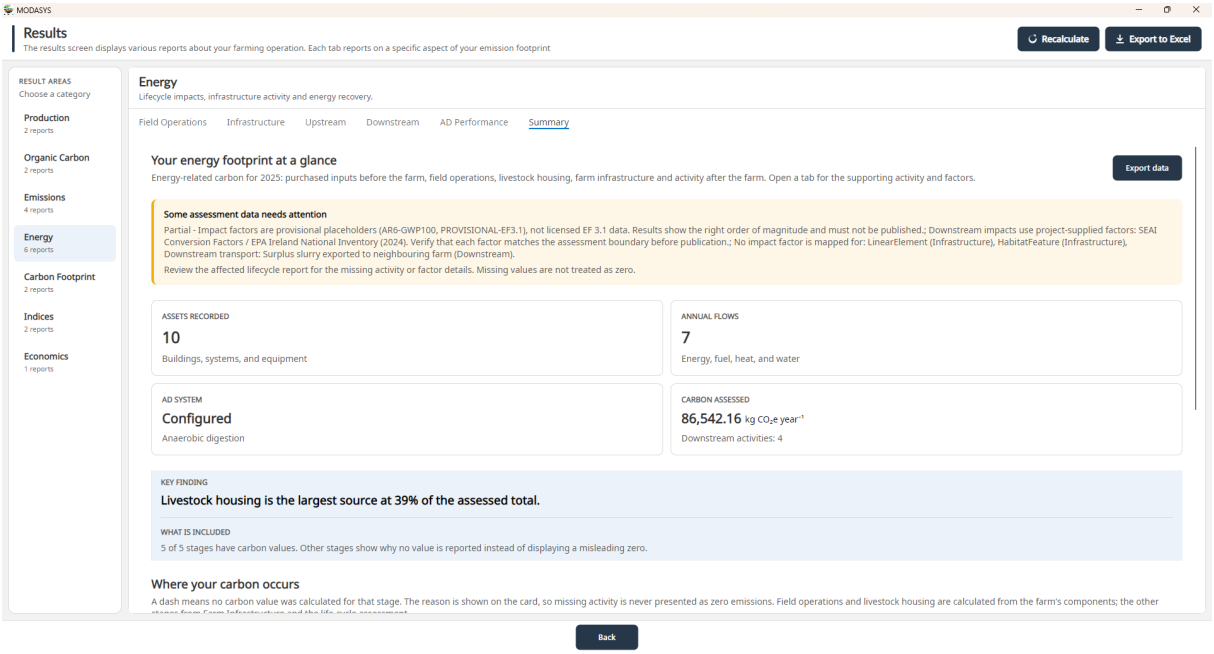



**Figure 7.16:** *Energy – Summary.*

## 7.5 Carbon Footprint

**Total Carbon Balance** For each land use (crop, grass, agroforestry, livestock) the chart shows every carbon flow on its own bar – soil organic carbon change, $N_2O$ and field energy – and the resulting total carbon balance, in kg $CO_2e$ (Figure 7.17). On the *Per hectare* basis each part is divided by its own land, and values across years are area-weighted. Upstream energy, including the making of a field's inputs (fertiliser, seed, sprays), is not on the fields: it is in the Farm energy bar of the Net Carbon Balance.

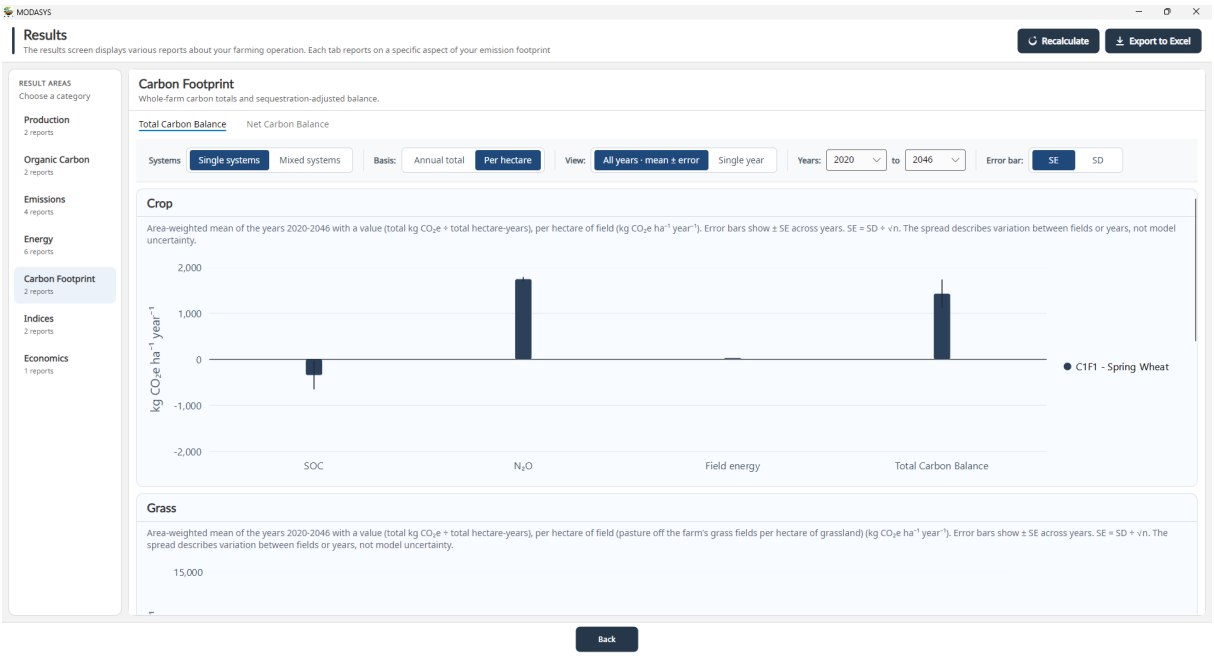



**Figure 7.17:** *Total Carbon Balance per hectare for the crop field. Further charts for grass, agroforestry and livestock follow below.*

**Net Carbon Balance** One bar per source – crop, grass, livestock, agroforestry and farm energy – each the source's net emissions (emissions minus removals), and a **Net Balance** bar that is the sum of the bars, in kg $CO_2$e (Figure 7.18). Farm energy holds the farm-level energy: the on-farm operation flows entered on the Energy tab, all upstream energy (field inputs, purchased farm inputs, infrastructure) and downstream energy (transport and processing), so the net balance counts the farm's energy from upstream to downstream. The energy of crop, grassland and livestock operations stays on the crop, grass and livestock bars. A net balance above zero means the farm is a net source of carbon; below zero, a net sink.

The **Systems** switch regroups the same results. *Single systems* shows each land use separately. *Mixed systems* groups the fields and herds that share land with each agroforestry system into one system named after the agroforestry type (e.g. Silvo Arable), with the rest shown as grass, livestock and farm energy (Figure 7.19). Every hectare is in exactly one system.

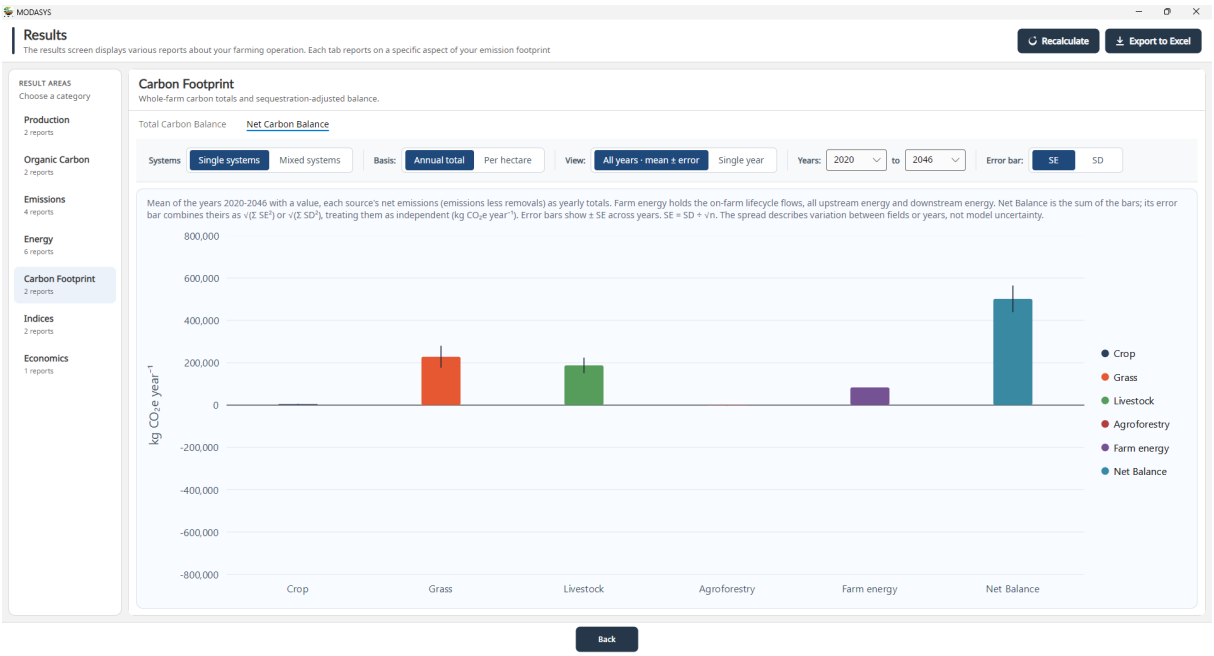



**Figure 7.18:** *Net Carbon Balance by source (single systems), mean of the years 2020–2046.*

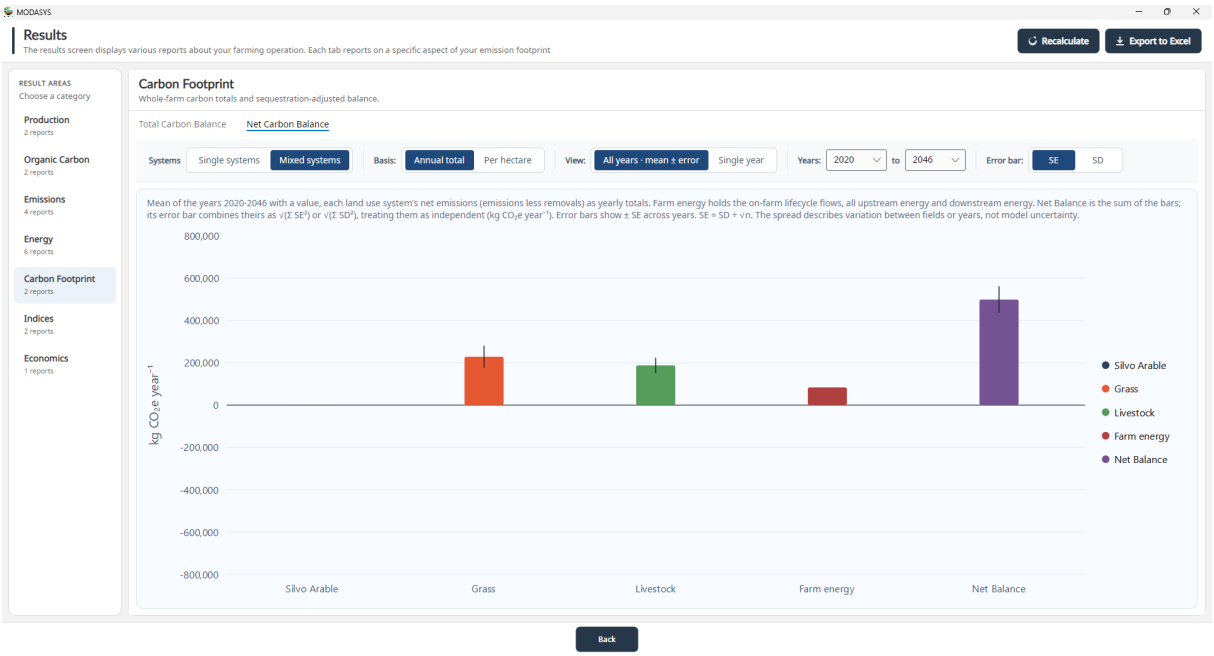



**Figure 7.19:** *Net Carbon Balance grouped as mixed systems.*

## 7.6 Indices

**Soil Health Index** compares each field's soil carbon with the reference for its land use and soil texture and reports the soil carbon ratio and a condition class from very low to very high (Figure 7.20). **Biodiversity Index** gives a continuous biodiversity score for each field and year: the land-use score, condition band, local and global biodiversity value and the contribution of each indicator (crop diversity, pesticide ecotoxicity, landscape features and so on) (Figure 7.21).

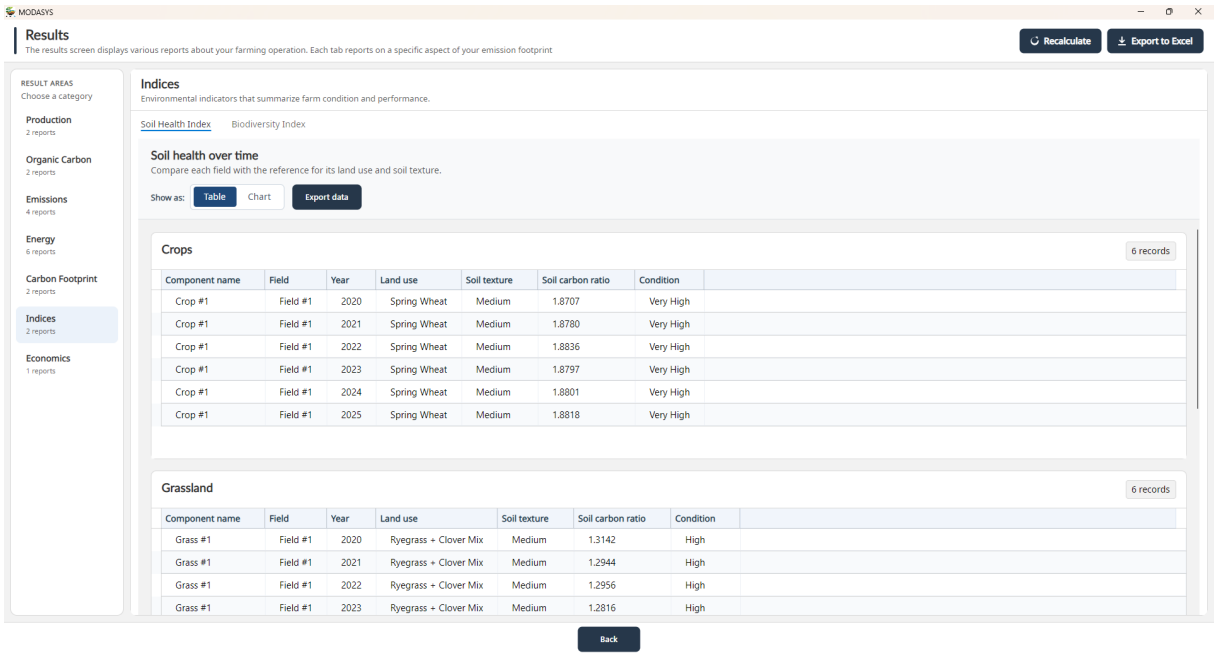



**Figure 7.20:** *Soil Health Index by field and year.*

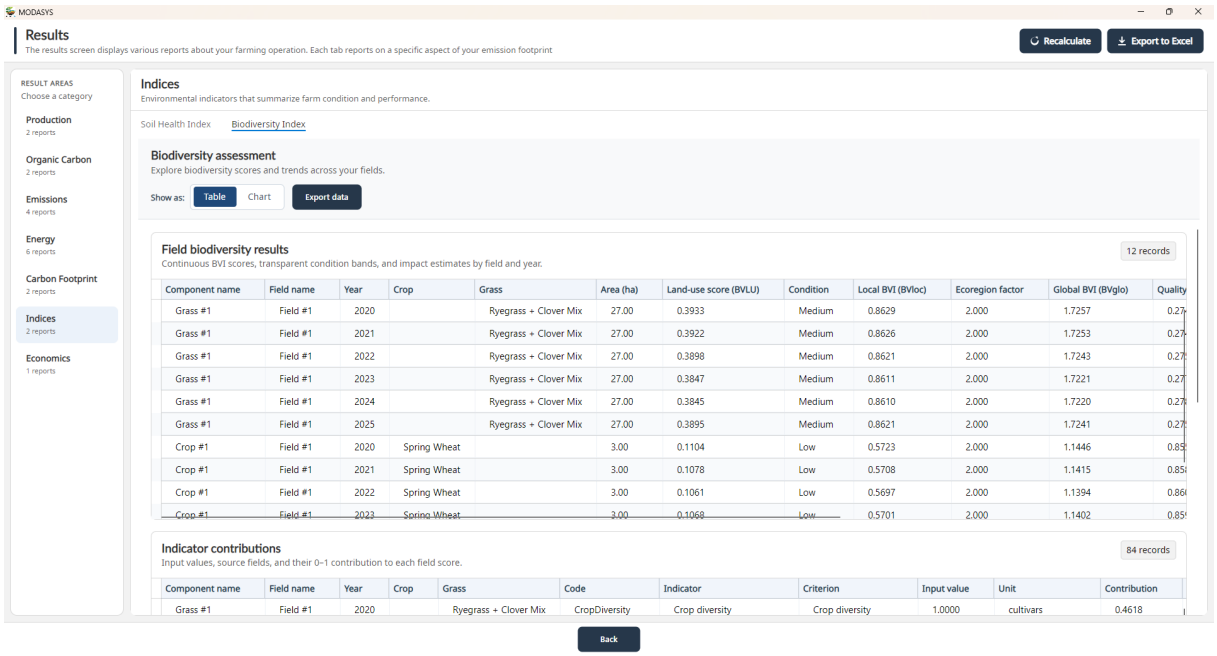



**Figure 7.21:** *Biodiversity Index: field scores (top) and indicator contributions (bottom).*

## 7.7 Economics

The Economics report shows production, price, revenue, fixed and variable costs and profit (€ $ha^{-1}$) for crops, grassland, livestock groups and agroforestry (Figure 7.22). **Show additional columns** adds the cost breakdown. Economic data come from several sources and give a rough estimate only.

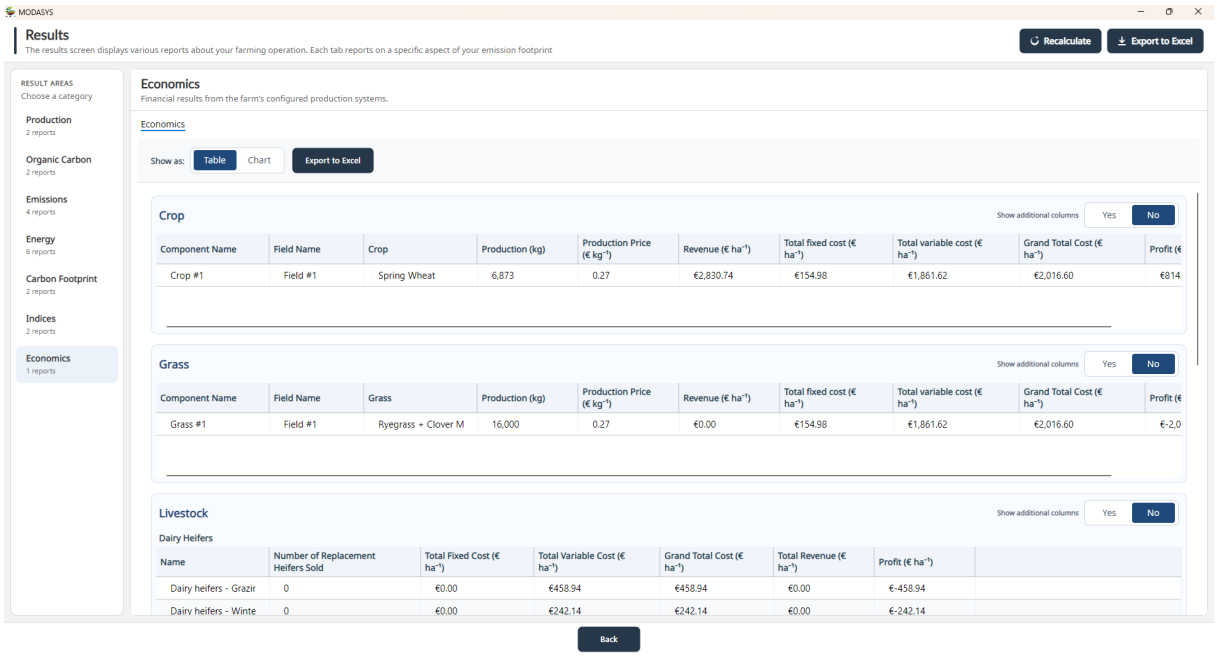



**Figure 7.22:** *Economics results.*

# 8. Conclusions

MODASYS builds on Holos V4.0 and HOLOS-IE to provide a whole-farm model for European agriculture. It brings together soil, crop, grass and livestock growth models, IPCC-based emission calculations, agroforestry carbon sequestration, life-cycle energy accounting, soil health and biodiversity indices and economics, so that the total and net carbon balance of a farm can be assessed in one place, for single components or for mixed systems. Through the HOLOSEU initiative, backed by European partners with funding from ERA-NET Co-Fund ICT-AGRI-FOOD, the platform is being extended from Ireland to other EU countries and climate regions, as shown on the region and country screen. Realising that potential depends on user feedback and on country-level validation and recalibration, which the UCD team and its partners are preparing. Active national and international collaboration will be vital to ensure the model's accuracy, credibility and relevance across diverse agricultural contexts.